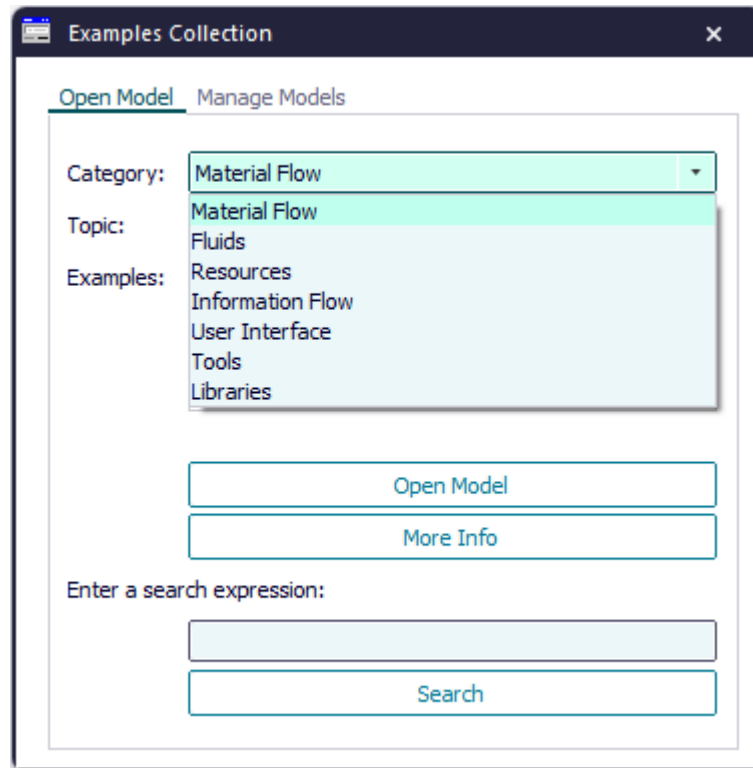




Go to [Use a Drag-and-Drop Control for Several Objects](#)

Back to [Work with Drag-and-Drop](#)

Modeling the Flow of Materials, Basics

Model the Flow of Materials, Basics

You can use a number of modeling techniques when creating your simulation models.

Below we introduce you to the basics of modeling the flow of materials in your simulation model. We'll show:

- How to create or introduce parts  into your model with the object *Source* .
- How to select and enter for how long a station  processes the parts that move onto it.
- How to set a station  up to process another type of part.
- How to transfer the parts from processing station  to processing station .
- How to model failures and failure times of the processing stations.
- How to remove the processed parts from your model with the object *Drain* .

Go to [Active and Passive Objects](#)

Go to [Produce Parts with the Source](#)

Go to [Produce and Process Parts with a Work Plan](#)

Go to [Remove Parts from the Installation with the Drain](#)

Go to [Transfer Parts from Station to Station](#)

Go to [Set a Station Up](#)

Go to [Define Processing Times](#)

Go to [Model Failures](#)

Go to [Buffer Parts within the Production Line](#)

Go to [Place Parts into Stock and Remove Parts from It](#)

Go to [Use the Store as Supermarket](#)

Go to [Stack Parts in the Store](#)

Go to [Loading a Container Layer by Layer](#)

Go to [Work with Workpiece Carriers](#)

Go to [Balance a Production Line](#)

Go to [Model the Flow of Materials, Advanced](#)

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Active and Passive Objects

Mobile and stationary material flow objects are the basic items you use when you create a simulation model.

- The **mobile material flow objects** *Part*, *Container* and *Transporter* represent the physical or logical parts moving through the model, i.e., your installation. These mobile objects (*MUs*) require active and passive material flow objects to process and transport them.
- The *Source* creates mobile objects at the beginning of the flow of materials through your plant. The *Drain* removes the *MUs* from the plant after the material flow objects processed them.

- The *FluidSource* creates *fluids* at the beginning of the flow of materials through your plant. The *FluidDrain* removes the fluids from the plant after the fluid objects processed them.
- The **active material flow objects** *Drain*, *Station*, *ParallelStation*, *AssemblyStation*, *DismantleStation*, *Conveyor*, *Sorter*, *PlaceBuffer*, and *Buffer* receive *MUs*, and process them for a certain time. They then attempt to **actively transfer the MUs** to the next object along material flow connections, symbolized by the *Connector*, **using the push-block principle**. They represent work stations in a factory, such as a lathe or a drilling station, etc. They only differ in the number of *MUs* they can process at the same time, a single one or several, and in the way they process them, in series or in parallel. Basically, the *Conveyor* resembles a conveyor system transporting *MUs* at a given speed over a given distance.
- The **active fluid objects** *FluidDrain*, *Tank*, *Mixer*, *Portioner*, and *DePortioner* receive *fluids*, and process them for a certain time. They then attempt to **actively transfer the fluids** to the next object along material flow connections, symbolized by the *Connector*, **using the push-block principle**.
- The **passive material flow objects** *Store*, *Track*, and *TwoLaneTrack* do not automatically transfer *MUs*. A *MU* remains in the *Store* until it is removed, for example by using a *Method*. You can only use the *Track* and the *TwoLaneTrack* in a meaningful way in conjunction with the *Transporter*, which drives along the *Track/TwoLaneTrack* with the speed you set. Model diverging strategies and converging strategies with the *FlowControl*.
- The **passive fluid objects** *Pipe* and *PatchMatrix* transport the *fluids* to their respective successor in the flow of materials.

We also distinguish between active and passive *MUs*:

- **Active MUs** are objects that can move by themselves, such as the *Transporter*, which can be self-propelled, and the *Worker*, who represents a human.
- **Passive MUs**, such as the *Part* and the *Container*, are objects that are transported from material flow object to material flow object, which process them. The *Part* represents any workpiece that is produced and processed. The *Container* represents any type of receptacle, such as a storage bin or a pallet.

In addition, we differentiate between:

- **Point-oriented objects** in which the *MUs* are located on a fixed processing place. For the *Source*, the *Drain*, the *Station*, the *ParallelStation*, the *AssemblyStation*, the *DismantleStation*, the *Sorter*, the *PlaceBuffer*, the *Buffer*, the *FluidSource*, the *FluidDrain*, the *Tank*, the *Mixer*, the *ContinuousMixer*, the *Portioner*, the *DePortioner*, and the *PatchMatrix*, their own real length and their dimension as well as the length and the dimension of the *MUs*, which transfer onto them, are irrelevant for the simulation.
- **Length-oriented objects** whose own real length and dimension as well as the length and the dimension of the *MUs*, which transfer onto them, are used during the simulation. Length-oriented objects are the *Conveyor*, the *Track*, the *TwoLaneTrack*, the *FootPath*, the *Pipe*, the *Container*, and the *Transporter*.

As opposed to the regular length-oriented objects, the *Turntable* consists of a single straight segment, not of a sequence of straight and curved segments. The *Turnplate* consists of a straight segment that sets the diameter of the *Turnplate*. The *AngularConverter* consists of two straight segments, not of a sequence of straight and curved segments. The *Converter* consists of a single straight segment.

The **length-oriented objects** are called **extrusion objects** in 3D.

Compare Work with Length-oriented Objects

Back to [Model the Flow of Materials, Basics](#)

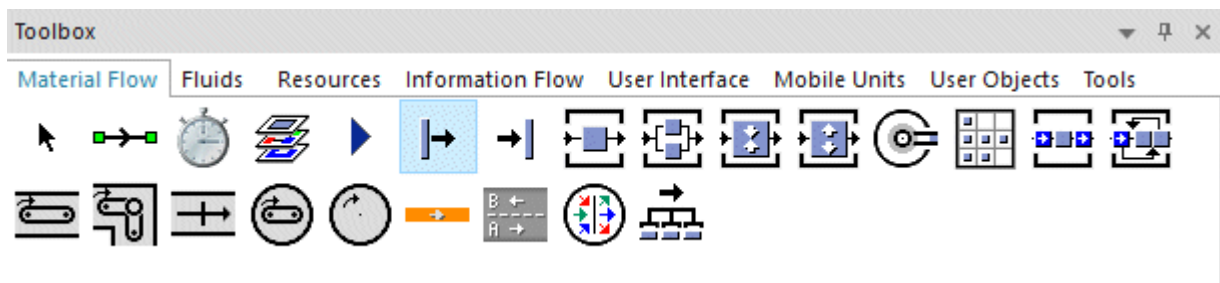
Producing Parts with the Source

Produce Parts with the Source

To create the *MUs*, which the material flow objects in your simulation model handle, we use the object *Source*. It might, for example, represent the *receiving* department of your installation that introduces parts produced at another location into the plant. Or it can be a machine, which produces the parts that the other stations process.

To remove these parts from the factory to model, for example, the *shipping* department, we use the object `Drain` →.

You can insert the **Source**  into your simulation model from the folder **MaterialFlow** in the *Class Library* or from the toolbar **Material Flow** in the *Toolbox*.



You can:

- **Select How the Source Proceeds, When it Cannot Produce MUs**
- **Produce Parts According to a Delivery Table**
- **Produce Parts During an Interval Which You Define**
- **Produce the Number of Parts You Need**
- **Produce Parts Using a Trigger Object**

Compare the sample models: Click the **Window** ribbon tab, click **Start Page > Getting Started > Example Models > Small Examples**. Then, select the respective **Category**, the **Topic**, and the **Example** in the dialog **Examples Collection**, and click **Open Model**.

Compare Check How Many Parts Were Introduced into the Plant

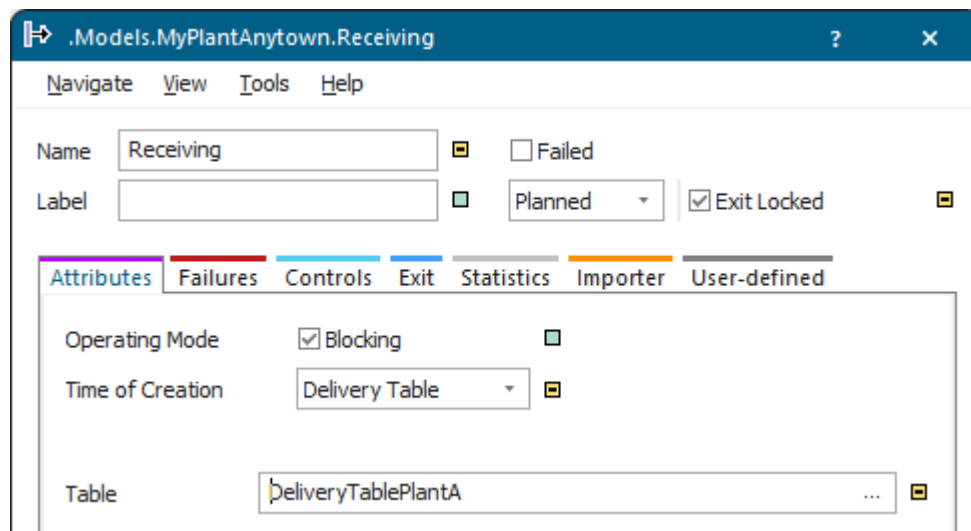
Compare Remove Parts from the Installation with the Drain

Back to [Model the Flow of Materials, Basics](#)

Select How the Source Proceeds, When it Cannot Produce MUs

Select how the *Source* proceeds when it cannot create the *MUs* at the **Time of Creation** that you entered because it cannot move the parts on to its successor. This is the case when it fails, when it is paused, or when it is blocked.

- Select the **Operating Mode** > **Blocking** ☒ **Blocking** , so that the *Source* remembers the time at which it was supposed to produce the next *MU* but could not. It then produces the following *MU* at the next possible point in time, i.e., if the *MU*, which blocked it, has moved on.



- Clear **Blocking**, so that the *Source* creates another *MU* exclusively at the time of creation you entered.

Note

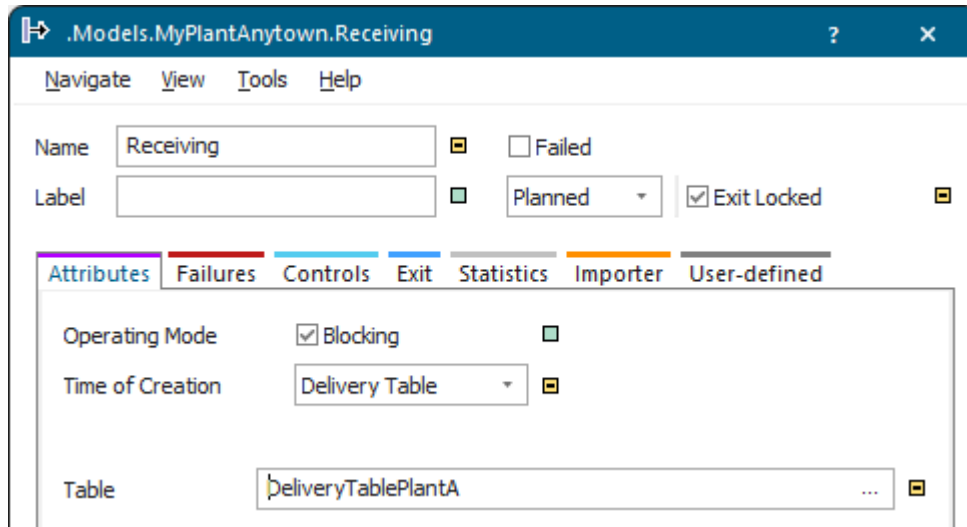
When the *Source* is temporarily not operational because it is **Failed**, **Paused**, or **Blocked**, the times of creation may shift, if you select **Blocking**. Then, the settings for the times of creation cannot be realized.

Back to [Produce Parts with the Source](#)

Produce Parts According to a Delivery Table

For the installation you are modeling you will, most likely, use a **Delivery Table** that contains the parts that a machine produces, parts provided by other departments within the same plant, or parts delivered from other plants.

Most of the time the delivery table you receive from another department will be a Microsoft Excel file containing, among others, the name of the part, its amount, its arrival time, etc. You will then open this Excel (*.xls) file in a *Plant Simulation* table  and save it there.



To produce *MUs* according to the time, the type, and the number of parts you typed into a *Delivery Table*:

- Select **Time of Creation** > **Delivery Table**.
- Click  and select the delivery table you would like to use in the dialog **Select Object**. Or
Drag your delivery table over the text box **Table** and drop it there.
- Click **OK**.

Next to information about the name and number of the produced parts, the *Delivery Table* may also contain values of their attributes. The *Delivery Table* has five columns.

Note

Instead of the data type *time* for the **Delivery Time** you can also use the data types *date*, *dateTime*, or *real*. When you use *date* or *dateTime*, the start time of the simulation has to be before the time you type in.

Each row of the *Delivery Table* defines a single order for producing *MUs*:

- Type the time at which the *Source* produces the *MUs* into the column **Delivery Time**.
- Type the class of the *MU* into the column **MU**. You can also use drag-and-drop of the *MU* class to accomplish this.
- Type the number of *MUs* to be produced into the column **Number**.

Note

When you type 0 as **Number** into a row, *Plant Simulation* does take the specified interval for the next creating cycle into account. This means that *Plant Simulation* does not skip this row, but does not produce a part during this cycle.

- Type a name for the *MUs* to be produced into the column **Name**.

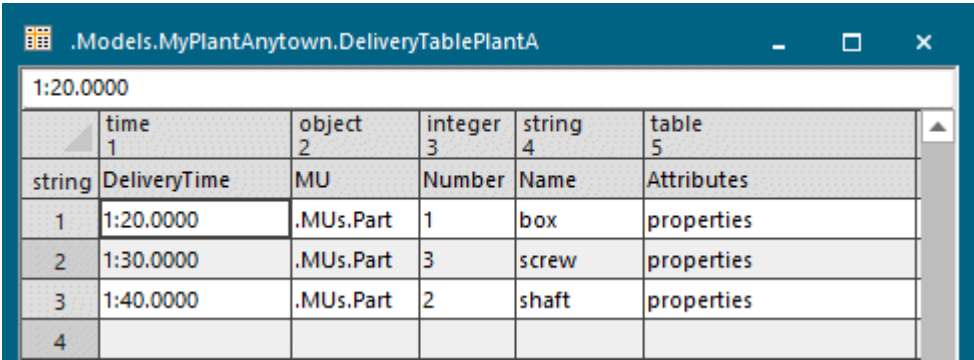
Note

You have to type in the **Delivery Time** and the **MU**.

If you do not type in a **Number**, *Plant Simulation* produces a single *MU*.

If you do not type in a **Name**, the *MUs*, which the *Source* produces, use the name of their class.

- In addition, you can, but you do not have to type the name of a subtable into the column **Attributes**. Type the attributes whose value the *Source* sets and *the user-defined attributes* that it creates, when it produces the *MUs*, into that subtable.



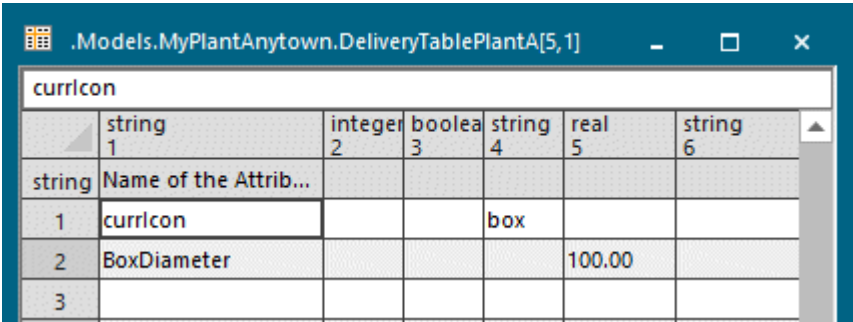
	time 1	object 2	integer 3	string 4	table 5
string	DeliveryTime	MU	Number	Name	Attributes
1	1:20.0000	.MUs.Part	1	box	properties
2	1:30.0000	.MUs.Part	3	screw	properties
3	1:40.0000	.MUs.Part	2	shaft	properties
4					

To open the subtable, do one of the following:

- Double-click the cell containing the subtable, **properties** in the column **Attributes** in the example above.
- Click into the cell and press **F2**.
- Right-click into the cell and select **Open Object**.

Type the name of a built-in attribute into the cells below column 1 of the attribute table. Type the value you want to assign to this attribute into one of the cells to the right with the respective data type. Or

Type the name of a *user-defined attribute* into the cells below column 1 of the table. If an attribute with that name does not exist yet, *Plant Simulation* creates it. Type the value you want to assign to this attribute into the cell to the right, which has the corresponding data type.



	string 1	integer 2	boolean 3	string 4	real 5	string 6
string	Name of the Attrib...					
1	currIcon			box		
2	BoxDiameter				100.00	
3						

Note

Plant Simulation assigns the data type of the column containing the value of the attribute to the data type of the attribute itself. *Plant Simulation* assigns the value you type into the cell to the generated attribute.

Back to [Produce Parts with the Source](#)

Producing Parts During an Interval Which You Define

Produce Parts During an Interval Which You Define

For the plant you are modeling you can produce parts during an interval which you define.

To produce *MUs* during the interval you define:

- Select **Time of Creation** > [Interval Adjustable](#).
- In our example, we chose to have the *Source* create one **Constant** type of *MU*, namely a **Part**, every minute and 20 seconds.
- Click **OK**.

.Models.MyPlantAnytown.Receiving

Navigate View Tools Help

Name: Receiving ☐ Failed

Label: ☐ Planned ☒ Exit Locked

Attributes Failures Controls Exit Statistics Importer User-defined

Operating Mode ☒ Blocking ☐

Time of Creation: Interval Adjustable ☐ Amount: -1 ☐

Interval: Const 1:20 ☐

Start: Const 0 ☐

Stop: Const 0 ☐

MU Selection: Constant ☐

MU: ~.~.MUs.Part ☐

3D OK Cancel Apply

The *Source*:

- Produces the first *MU* at the time you enter for **Start**.
- Produces the next *MU* after the time you enter for **Interval** has elapsed.
- Stops producing *MUs* at the point in time you enter for **Stop**. Enter 0 into the text box **Stop** if you do not want to apply a time limit.

When you select **Time of Creation > Interval Adjustable** or **Time of Creation > Number Adjustable** you can also select how the *Source* produces which types of *MUs*.

You can:

- **Produce a Single Part Type Only**
- **Produce Parts in a Fixed Sequence Over and Over Again**
- **Produce Parts in a Fixed Sequence One Time Only**
- **Produce Parts With a Random Frequency Entered into a Data Table**
- **Produce Parts With a Percentage Entered into a Data Table**

Compare **Specify Times in Object Dialogs**

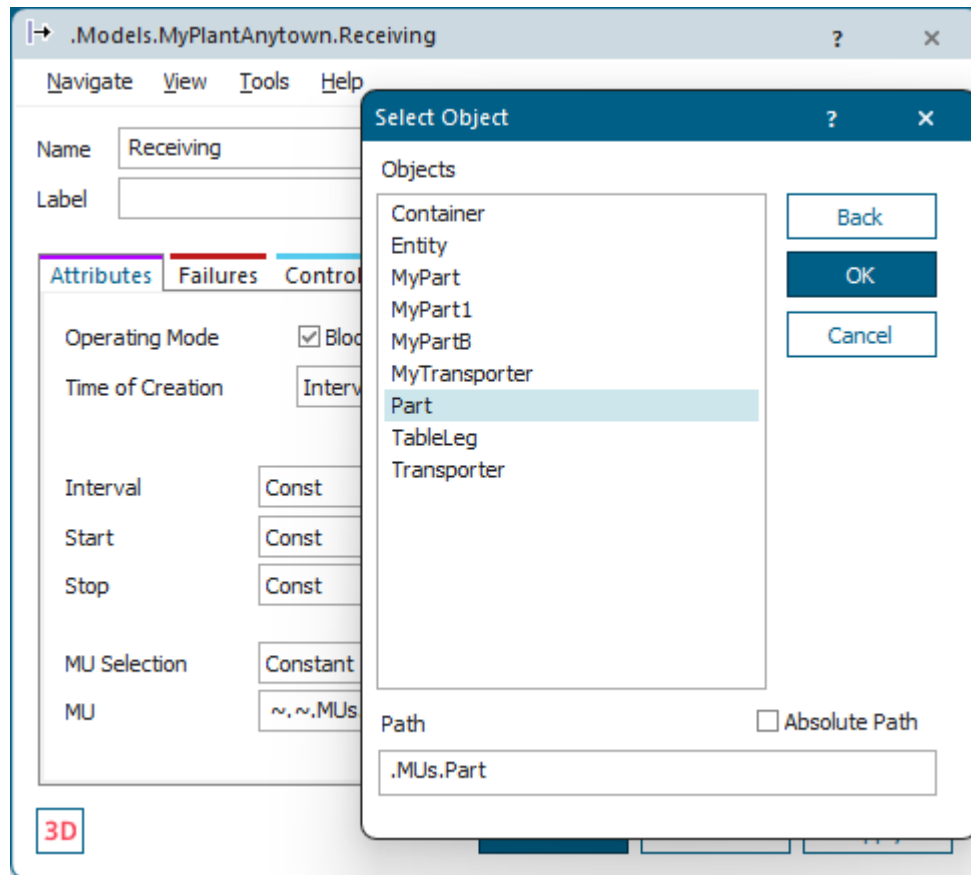
Back to **Produce Parts with the Source**

Produce a Single Part Type Only

For the plant you are modeling you can produce a single part type only.

To make the *Source* produce one type of *MU* only:

- Select **MU Selection > Constant**.
- Click . In the dialog **Select Object** navigate to the folder where the *MUs* are located. Select the name of a **MU** class and click **OK**.



Back to Produce Parts During an Interval Which You Define

Produce Parts in a Fixed Sequence Over and Over Again

For the plant you are modeling you can produce parts in a fixed sequence over and over again.

To make the *Source* produce *MUs* in a fixed sequence over and over again:

- Select **MU Selection** > **Sequence Cyclical**.
- Insert a table into a *Frame* or into a folder in the *Class Library*.
- To designate the *MU* class to be produced, do one of the following:
 - Click . In the dialog **Select Object** navigate to the folder where the data table is located into which you want to enter the *MUs*. Select the name of the **Table** and click **OK**.
 - Drag the table over the text box **Table** and drop it there.
 - Enter the name of and the path to the table into the text box next to **Table**.

Name: ☐ Failed
 Label: ☐ Planned ☐ Exit Locked

Attributes Failures Controls Exit Statistics Importer User-defined

Operating Mode ☒ Blocking ☐
 Time of Creation ☐ Amount ☐
 Interval ☐
 Start ☐
 Stop ☐
 MU Selection ☐ ☐ Generate as Batch ☐
 Table ☐

.Models.TableLeg				
	object 1	integer 2	string 3	table 4
string MU	Number	Name	Attributes	
1	.Models.TableLeg	8	Leg_A	a1
2	.Models.TableLeg	8	Leg_B	a2
3				

.Models.TableAssembly.TableLegs[4, 1]					
	string 1	integer 2	boolean 3	string 4	real 5
string Name of Attribute					
1	PictureName3D			Leg_A	
2					
3					

- Enter the names of the *MU* classes to be produced into the cells below **MU** of the table.
- You can, but you do not have to enter a **Name** and **Attributes** into columns 3 and 4 of the table.
- Enter the number of *MUs* to be produced into the cells below **Number**.

Note

When you do not enter a **Name**, the *MUs* that the *Source* produces use the name of their class.

- To make the *Source* produce the number of *MUs* you specified in a row of the sequence table all at once in a single batch before moving the entire batch on, select **Generate as Batch**. Once the *Source* has processed the entire sequence, it starts processing the information in the table again starting at the beginning of the sequence.

To make the *Source* produce the *MUs* is a sequence of individual *MUs*, clear the check box. When the *Source* produces the *MUs* is a batch, it attempts to produce the entire set of Parts at the given start time and to move them on to the next object.

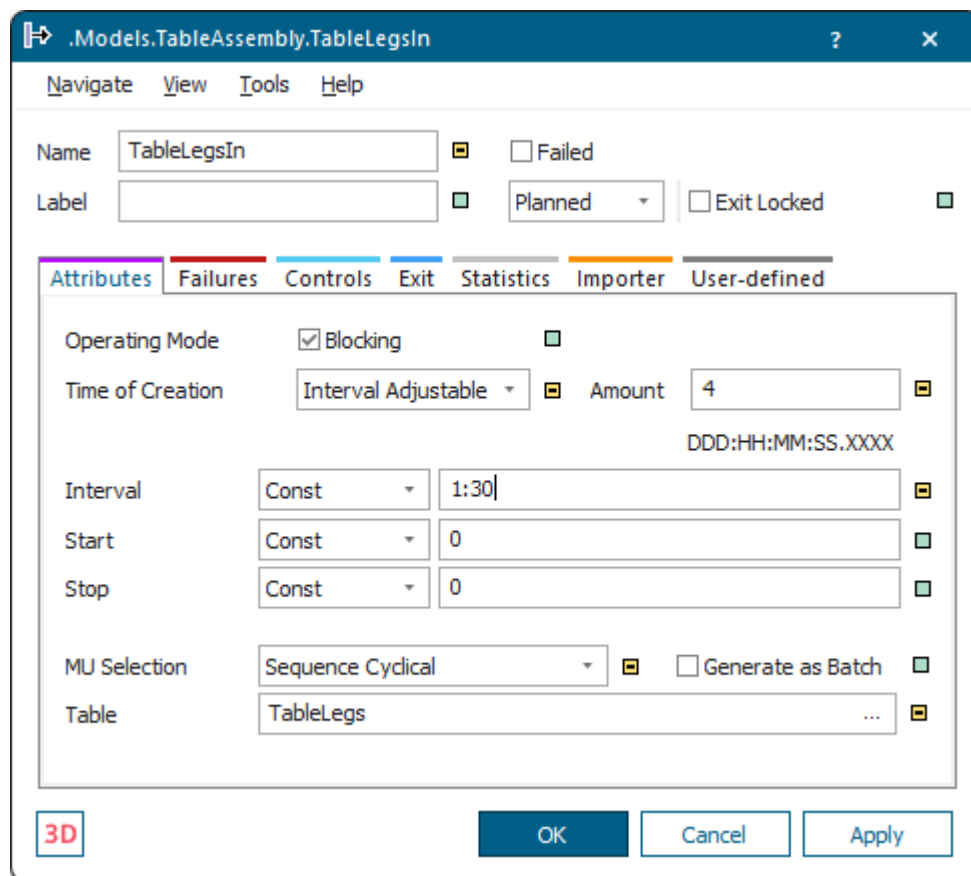
Back to Produce Parts During an Interval Which You Define

Produce Parts in a Fixed Sequence One Time Only

For the plant you are modeling you can produce parts in a fixed sequence one time only.

To make the *Source* produce *MUs* in a fixed sequence one time only:

- Select **MU Selection** > **Sequence Cyclical**.

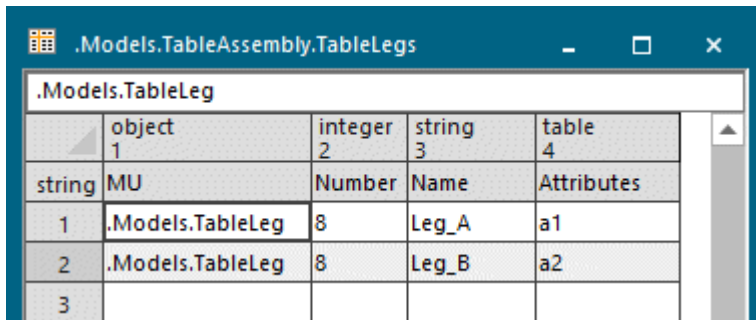


Note

The setting **Sequence** is not available, when you set **Time of Generation** to **Number Adjustable**.

- Insert a table into a *Frame* or into a folder in the *Class Library*.
- To designate the type of *MU* to be produced, do one of the following:
 - Click . In the dialog **Select Object** navigate to the folder where the data table is located into which you enter the *MUs*. Select the name of the **Table** and click **OK**.

- Drag the table over the text box **Table** and drop it there.
- Enter the name of and the path to the table into the text box next to **Table**.



	object 1	integer 2	string 3	table 4	
string	MU	Number	Name	Attributes	
1	.Models.TableLeg	8	Leg_A	a1	
2	.Models.TableLeg	8	Leg_B	a2	
3					

- Enter the names of the *MU* classes to be produced into the cells below **MU** into the table.
- Enter the number of *MUs* to be produced into the cells below **Number**.
- You can, but you do not have to enter a **Name** and **Attributes** into columns 3 and 4 of the table.

Note

When you do not enter a **Name**, the *MUs* that the *Source* produces use the name of their class.

- To make the *Source* produce the number of *MUs* you specified in a row of the sequence table all at once in a single batch before moving the entire batch on, select **Generate as batch**.

To make the *Source* produce the *MUs* as a sequence of individual *MUs*, clear the check box. When the *Source* produces the *MUs* as a batch, it attempts to produce the entire set of parts at the given start time and to move them on to the next object all at once in the batch.

Back to Produce Parts During an Interval Which You Define

Produce Parts With a Random Frequency Entered into a Data Table

For the plant you are modeling you can produce parts with a random frequency entered into a data table.

To make the *Source* produce *MUs* with a frequency, which you type into a table:

- Select **MU Selection** > **Random**.

.Models.TableAssembly.TableLegsIn

Name: TableLegsIn ☐ Failed
 Label: ☐ Planned ☐ Exit Locked

Attributes Failures Controls Exit Statistics Importer User-defined

Operating Mode: ☒ Blocking ☐
 Time of Creation: Interval Adjustable ☐ Amount: -1 ☐
 DDD:HH:MM:SS.XXXX
 Interval: Const 1:00 ☐
 Start: Const 0 ☐
 Stop: Const 0 ☐
 MU Selection: Random ☐ ☐ Generate as Batch ☐
 Table: TableTopsRandom ☐

3D OK Cancel Apply

- Insert a table into a *Frame* or into a folder in the *Class Library*.
- To designate the type of *MU* to be produced, do one of the following:
 - Click . In the dialog **Select Object** navigate to the folder where the data table is located into which you enter the *MUs*. Select the name of the **Table** and click **OK**.
 - Drag the table over the text box **Table** and drop it there.
 - Enter the name of and the path to the table into the text box next to **Table**.

	object 1	real 2	integer 3	string 4	table 5
string	MU	Frequency	Number	Name	Attributes
1	.Models.TableTop	1.00	6	TableTop_A	a1
2	.Models.TableTop	2.00	10	a2	a2
3					

- Enter the names of the *MU* classes to be produced into the cells below **MU** into the table.
- Enter the frequency number for each class of *MU* to be produced into the cells below **Frequency**.

When the *Source* produces the *MUs*, *Plant Simulation* selects and processes a random generation order according to the frequency you entered here.

- Enter the number of *MUs* to be produced into the column **Number**.
- You can, but you do not have to enter a **Name** or an **Attributes** table.

Note

If you do not enter a **Name**, the *MUs* that the *Source* produces use the name of their class.

To check which part types were produced at which point in time, you can activate the check box **Creation table** on the tab **Statistics** and then open the table.

Back to [Produce Parts During an Interval Which You Define](#)

Produce Parts With a Percentage Entered into a Data Table

For the plant you are modeling you can produce parts with a percentage entered into a *DataTable*.

To make the *Source* produce *MUs* with a frequency, which you type into a table:

- Select **MU Selection** > [Percentage](#).

.Models.MyPlantAnytown.Receiving

Navigate View Tools Help

Name: Receiving ☐ Failed ☐ Label: ☐ Planned ☒ Exit Locked ☐

Attributes Failures Controls Exit Statistics Importer User-defined

Operating Mode ☒ Blocking ☐

Time of Creation: Interval Adjustable ☐ Amount: -1 ☐

Interval: Const ☐ 1:20 ☐

Start: Const ☐ 0 ☐

Stop: Const ☐ 0 ☐

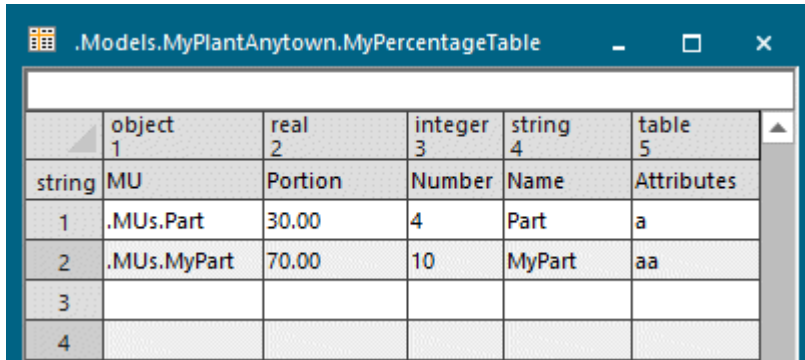
MU Selection: Percentage ☐ ☒ Generate as Batch ☐

Table: MyPercentageTable ☐

3D OK Cancel Apply

- Insert a table into a *Frame* or into a folder in the *Class Library*.
- To designate the type of *MU* to be produced, do one of the following:

- Click . In the dialog **Select Object** navigate to the folder where the data table is located into which you enter the *MUs*. Select the name of the **Table** and click **OK**.
- Drag the table over the text box **Table** and drop it there.
- Enter the name of and the path to the table into the text box next to **Table**.



	object 1	real 2	integer 3	string 4	table 5
string	MU	Portion	Number	Name	Attributes
1	.MUs.Part	30.00	4	Part	a
2	.MUs.MyPart	70.00	10	MyPart	aa
3					
4					

- Enter the names of the *MU* classes to be produced into the cells below **MU** into the table.
- Enter the percentage portion for each class of *MU* to be produced into the cells below **Portion**.
- Enter the number of parts to be produced into the cells below **Number**.
- You can, but you do not have to enter a **Name** or an **Attributes** table.

Note

If you do not enter a **Name**, the *MUs* that the *Source* produces use the name of their class.

Compare **Specify Times in Object Dialogs**

Back to **Produce Parts with the Source**

Video on YouTube

https://youtu.be/BvHLQCljGIA?si=b7Ua-NI7_uxT39m4&t=158

Produce the Number of Parts You Need

For the plant you are modeling you can produce the number of parts you need.

To produce a certain number of *MUs* during the interval you define:

- Select **Time of Creation** > **Number Adjustable**.

The screenshot shows the **.Models.MyPlant.Source** dialog box. The **Attributes** tab is selected. The **Name** field is set to **Source**. The **Label** field is empty. The **Operating Mode** is set to **Blocking** (checked). The **Time of Creation** is set to **Number Adjustable**. The **Amount** is set to **1200**. The **Creation times** are set to **Normal** with values **0:08, 1:40**. The **MU Selection** is set to **Constant**. The **MU** field contains **MUs.Part**. There are checkboxes for **Failed**, **Planned**, and **Exit Locked**. At the bottom, there are buttons for **3D**, **OK**, **Cancel**, and **Apply**.

- Enter the **Amount**, i.e., the number of *MUs* to be created.
- Select the **Creation Times**, which determine the points in time at which the *Source* creates the *MUs*.
Select a distribution from the drop-down list **Creation Times** and enter the values that the distribution requires into the text box.

The *Source* produces the number of *MUs* at the different times which the random number generator generated at the beginning of the simulation. The determined times are contained within the bounds if you enter a lower bound and an upper bound.

If you select a **Constant** time, the *Source* creates the number of *MUs* all at one point in time.

Note

If you set the **Creation Times** via *SimTalk*, use the attribute **Interval**.

Note

For the setting **Number Adjustable** you cannot select the **MU Selection** > **Sequence**.

- Click **OK**.

If you select **Time of Creation** > **Interval Adjustable** or **Time of Creation** > **Number Adjustable** you can also select how the *Source* produces which types of *MUs*.

Compare Specify Times in Object Dialogs

Back to Produce Parts with the Source

Produce Parts Using a Trigger Object

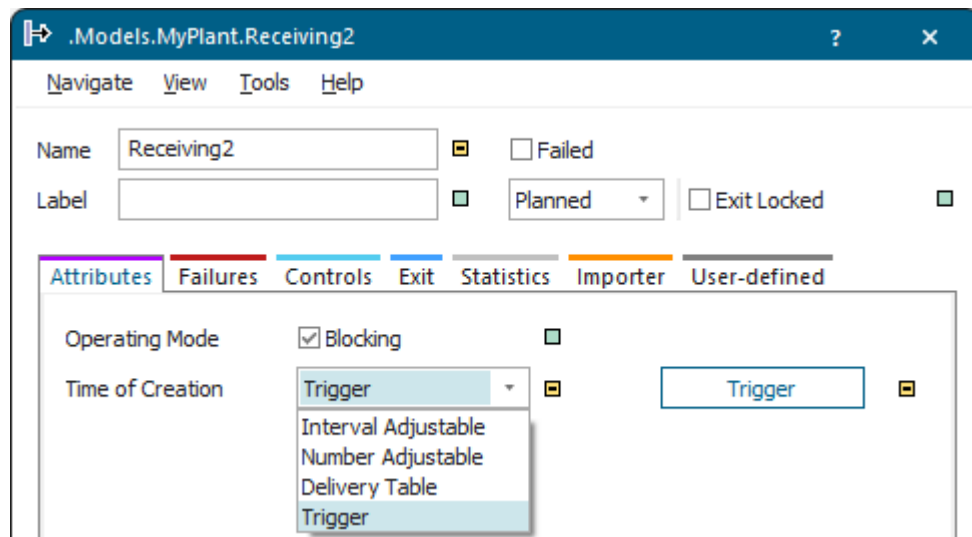
For the plant you are modeling you can produce parts using a *Trigger*.

You might, for example, use a *Source* that is controlled by a *Trigger*:

- To set a time of creation, such as with a delivery table, and
- To periodically repeat using the values in this table, for example when you select **Sequence Cyclical** from the drop-down list **MU Selection**.

Proceed as follows:

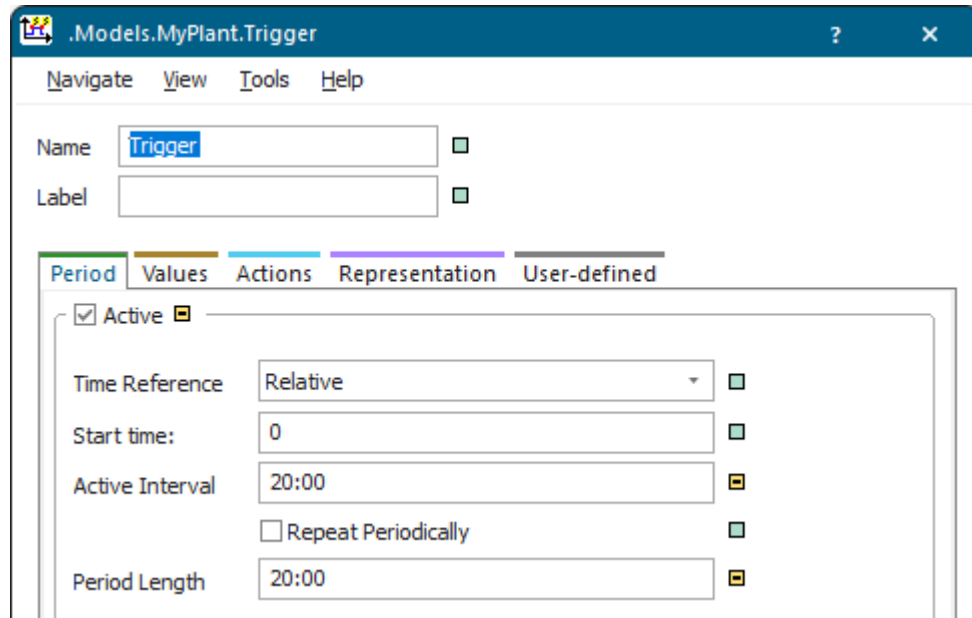
- Select **Time of Creation** > **Trigger** in the dialog of the *Source*.



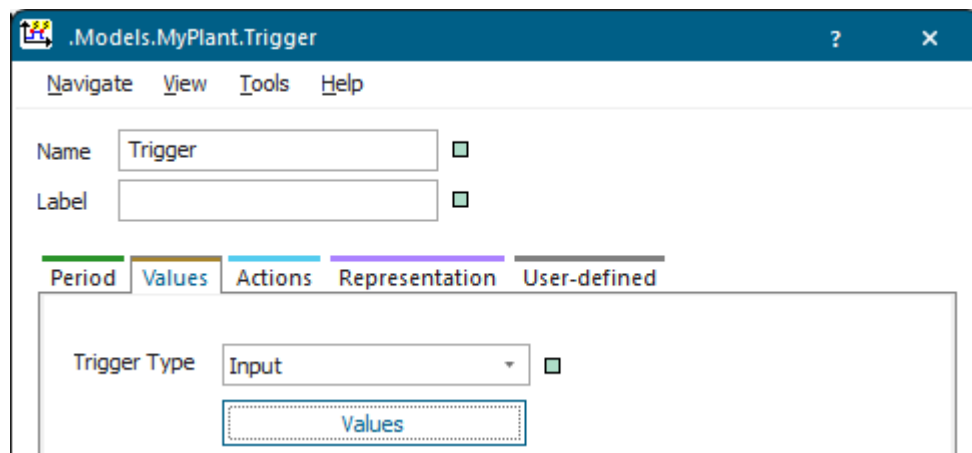
Before you can enter the *Trigger*, click the check box **Inheritance**, so that it looks like this .

- Click **Trigger**, drag the *Trigger*  that controls the *Source* from the *Frame* over the list that opens and drop it there.
- In the dialog of the *Trigger* enter the **Active Interval**, i.e., the interval during which the *Trigger* will be active.

Enter the **Period Length**, i.e., the duration of the *Trigger's* cycle.



- Click the tab **Values** and select the **Trigger Type > Input**.



- Click the button **Values** and enter into the *TimeSequence* object that opens:

The **Point in Time** at which the *Source* creates *MUs* into the cells on the left-hand side.

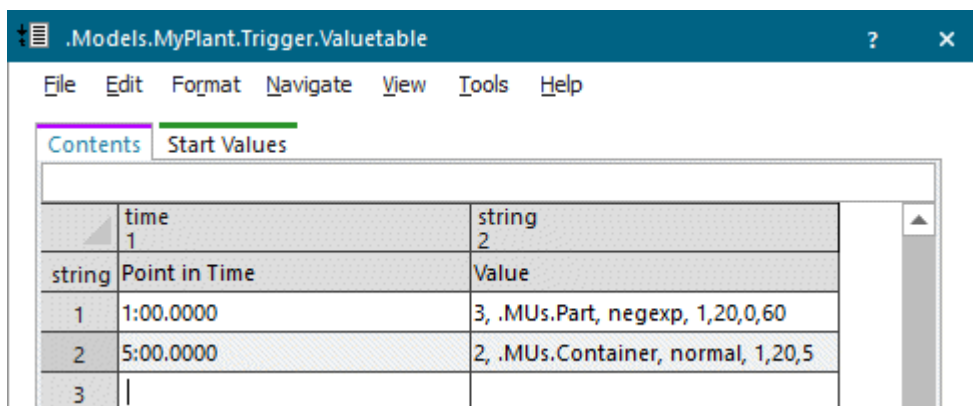
The current sequence of **Values** into the cells on the right-hand side. An order is a *string* with this format: amount,mu_Type,distributionType[,distribution parameters].

Note

The string defining this sequence of values may not contain any blank spaces.

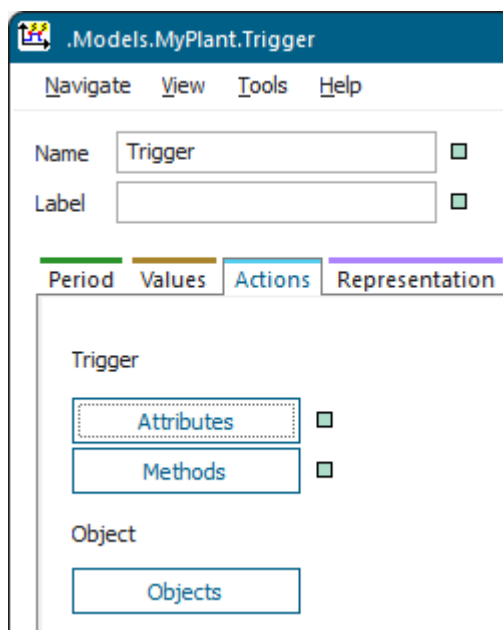
You have to enter the amount of *MUs* to be produced, the type to be produced, and at least a constant value. If you enter just *Const*, the *Source* produces the *MUs* at the point in time, which you typed into the cell to the left. If you would like it to produce the *MUs* with an offset to the time you entered there, enter the number of seconds after which it produces them after *Const*.

If you enter a distribution, its values set the time offset to the time you entered into the corresponding left cell. The offset you enter has to be a positive number!



	time	string
1	1:00.0000	3, .MUs.Part, negexp, 1,20,0,60
2	5:00.0000	2, .MUs.Container, normal, 1,20,5
3		

- Click the tab **Actions** and click **Objects**. The *Trigger* shows the *Source* in the table.



Name: ☐

Label: ☐

Period Values **Actions** Representation

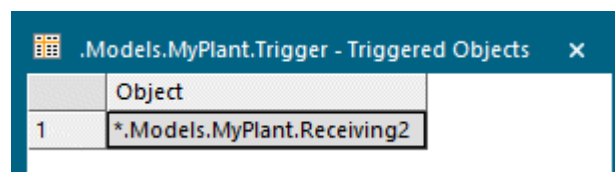
Trigger

Attributes ☐

Methods ☐

Object

Objects ☐



	Object
1	*,Models.MyPlant.Receiving2

Compare [Check How Many Parts Were Introduced into the Plant](#)

Back to [Produce Parts with the Source](#)

Producing and Processing Parts with a Work Plan

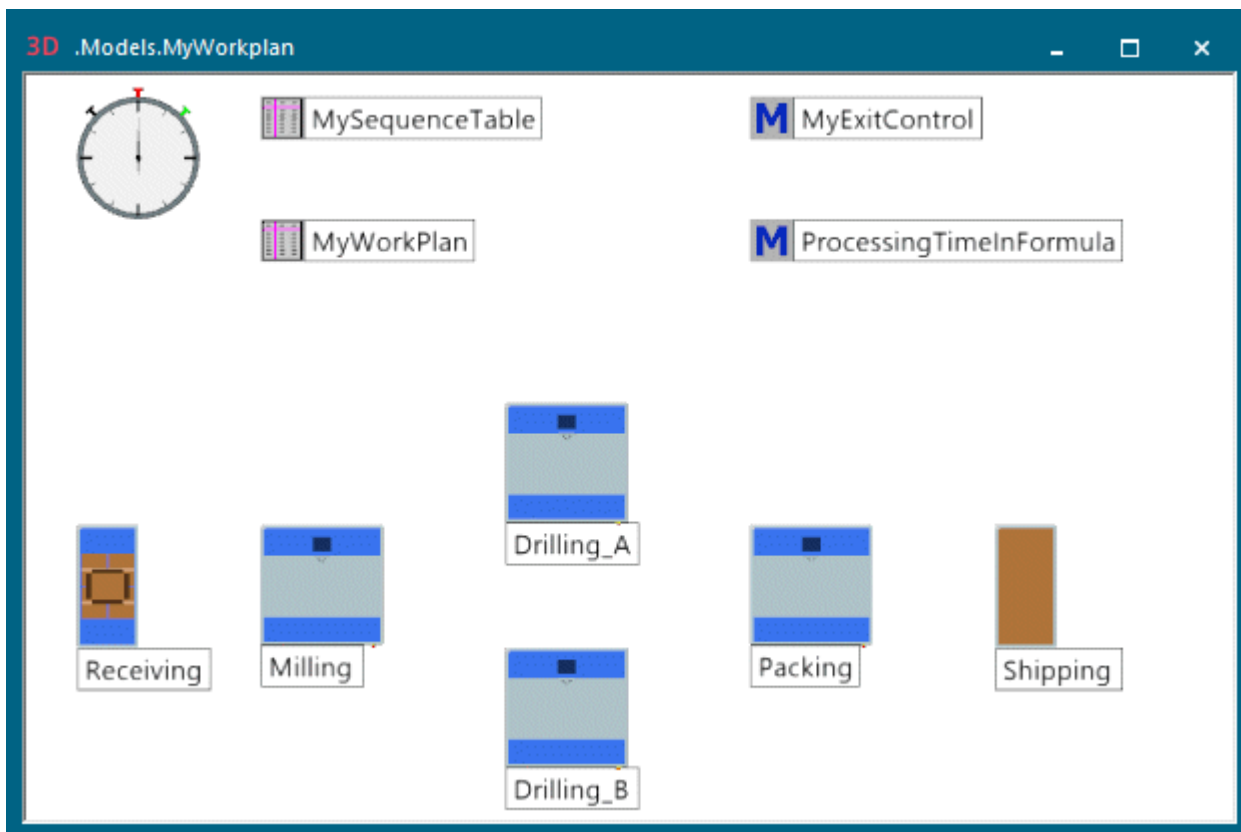
Produce and Process Parts with a Work Plan

A **work plan** or operations plan determines how a product is produced. It lists the various production steps in the sequence in which they are executed.

Note

We neither specify the cost center, to which each of the individual production steps is charged in *Plant Simulation*, nor the allowed time for each step. This you can do in *Process Designer*.

For simplicity's sake each operation can only be executed by a single station in our example. Our work plan thus is described by a sequence of stations. The *Source* named *Receiving* holds position **0** in the work plan, the station *Milling* position **1**, the stations *Drilling_A* or *Drilling_B* position **2** and the station *Packing* position **3**.



To facilitate quick and easy modeling:

- We create the work plan in a *DataTable*  and insert it into the *Frame*, in which we create our simulation model. We enter the set-up times of the stations and the processing times of the different parts into sub-tables of the work plan.
- We produce two different types of parts with a *Source*  using a sequence table. The parts need two *user-defined attributes*. One sets the name of the part. The other sets the position of the part in the work plan.
- We program a *Method* , i.e., an **Exit Control**, which enables the processing station to find the next station in the sequence of operations after each processing step and moves the part on to this station. We insert this *Method*, which all stations access, into the *Frame*.

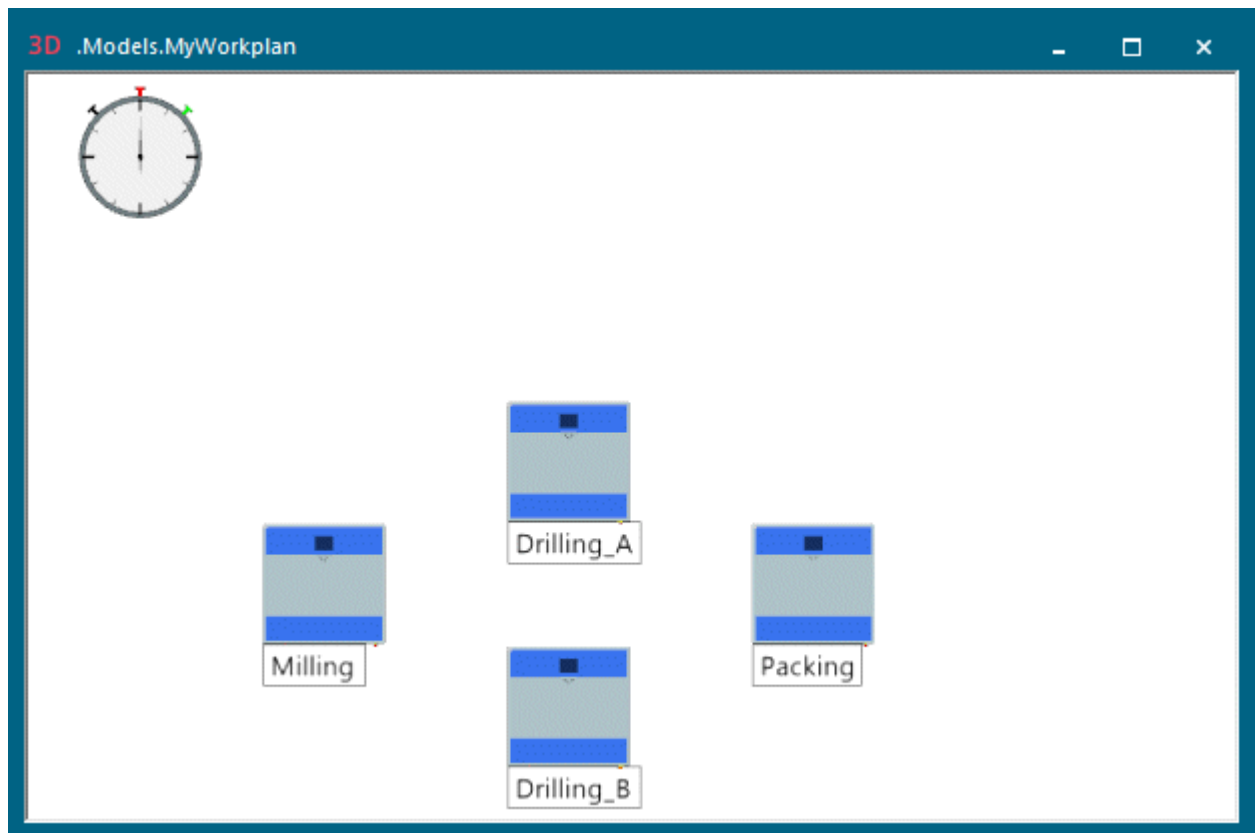
Back to [Model the Flow of Materials, Basics](#)

Creating the Processing Stations

Create the Processing Stations

To be able to assign the operations and the stations in the work plan later on, we start out with creating the stations which execute the operations.

- To create the class of the processing station, right-click the object *Station* in the folder **MaterialFlow** and select **Derive** on the context menu.
- To move the derived *Station* to the folder in which we are going to create the model, hold down **Shift** and drag it so that folder. Rename it to *MyStation*. By default *Plant Simulation* places derived objects into the folder **UserObjects**.
- Insert it from there into the model three times. Rename the stations so that they match the names of the operations. In our example we named them *Milling*, *Drilling_A*, *Drilling_B* and *Packing*.



As the processing stations share a number of features, we set these in their class, *MyStation* in our example. The class object passes these properties on to its instances.

We will:

- Define Times in the Class of the Processing Stations
- Define Set-up Behavior in the Class of the Processing Stations
- Specify the Name of the Exit Control in the Class of the Processing Stations

Back to Produce and Process Parts with a Work Plan

Define Times in the Class of the Processing Stations

We define the **Processing Time** and the **Set-up Time** in a **Formula** in the class of our processing stations.

- We want the stations to get the **Processing Time** of all parts from the work plan *MyWorkPlan*. This *DataTable* is located in the *Frame* in which we build the model, i.e., the *root Frame*. The station then opens the subtable **Operations** for the respective part and gets the times in the column **Processing Time** of the respective station within this subtable.

The first screenshot shows the main table with columns: string 0, string Entity type, table 1, and Operations. It lists operations for MyPart A and MyPart B.

string 0	string Entity type	table 1	Operations
1	MyPart A		Operations for MyPart A
2	MyPart B		Operations for MyPart B
3			
4			

The second screenshot shows a subtable titled 'Operations for MyPart B' with columns: string 0, string Entity type, table 1, and Operations. It lists operations for MyPart A and MyPart B.

string 0	string Entity type	table 1	Operations
1	MyPart A		Operations for MyPart A
2	MyPart B		Operations for MyPart B
3			
4			

The third screenshot shows a subtable titled '.Models.MyWorkplan.MyWorkPlan[1,2]' with columns: object 0, time 1, and time 2. It lists operations for MyPart A and MyPart B.

object 0	time 1	time 2
1	10.0000	1:00.0000
2	20.0000	2:00.0000
3	10.0000	3:00.0000
4		

For the **Processing Time** we use a *Method* which we call from the formula. The *Method* is called *ProcessingTimeInFormula*.

The source code looks like this:

```
-> time
var PartType := @.EntityType
var Station  := ?
var WorkPlan := root.myWorkPlan["Operations", PartType]
result      := WorkPlan["Processing time", Station]
```

```

M .Models.MyWorkplan.ProcessingTimeInFormula
-> time
var PartType := @.EntityType
var Station := ?
var WorkPlan := root.myWorkPlan["Operations", PartType]
result := WorkPlan["Processing time", Station]

```

- We want the stations to get the **Set-up Time** of all parts from the work plan *MyWorkPlan*. This *DataTable* is located in the *Frame* in which we build the model, i.e., the *root Frame*. The station then opens the subtable **Operations** for the respective part and gets the times in the column **Set-up Time** of the respective station within this subtable. *Self* identifies the station contained in the row on the subtable.

The description above translates to this formula, which we directly type into the text box:

```
root.MyWorkPlan["Operations",@.EntityType]["Setup time",Self]
```

The screenshot shows the configuration window for a station named 'MyStation'. The 'Times' tab is selected, displaying various time-related settings. The 'Set-up Time' is configured using a formula that references the 'Operations' subtable of the 'MyWorkPlan'.

Field	Type	Value
Processing Time	Formula	ProcessingTimeInFormula
Set-up Time	Formula	root.MyWorkPlan["Operations",@.EntityType]
Recovery Time	Const	0
Recovery Time Starts	When part enters	
Cycle Time	Const	0

Back to [Create the Processing Stations](#)

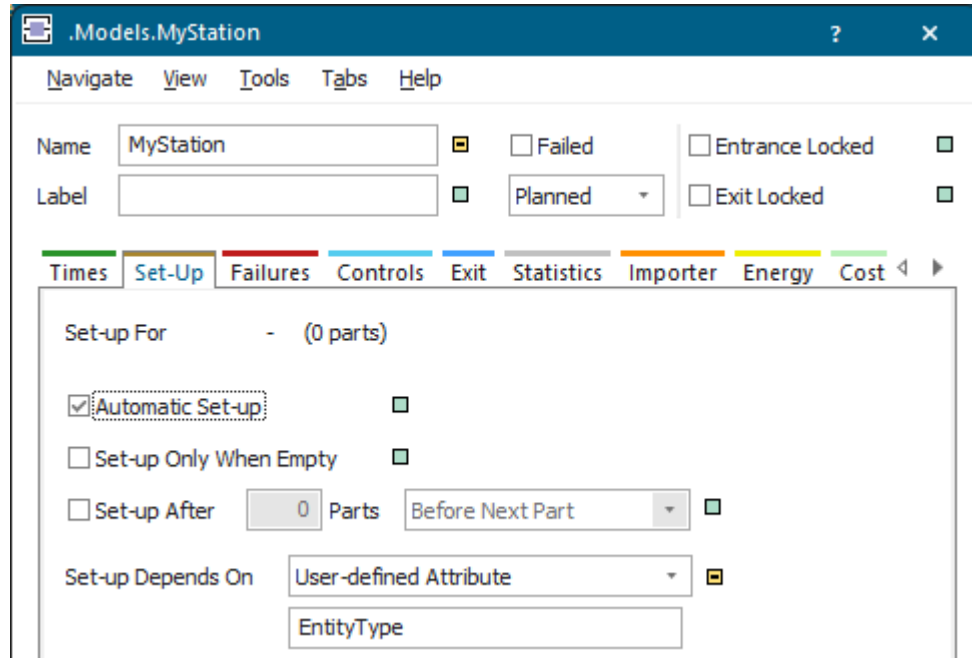
Define Set-up Behavior in the Class of the Processing Stations

We

To automatically set the stations up depending on the name of a user-defined attribute of the *MU*:

- Select the check box **Automatic**.

- Select **User-defined Attribute** from the drop-down list **Set-up depends on**. Enter the name of the *user-defined attribute* which we defined in the **Sequence table** with which the *Source* produces parts, compare **Produce the Parts with a Source Using a Sequence Table**. We typed in `EntityType`.

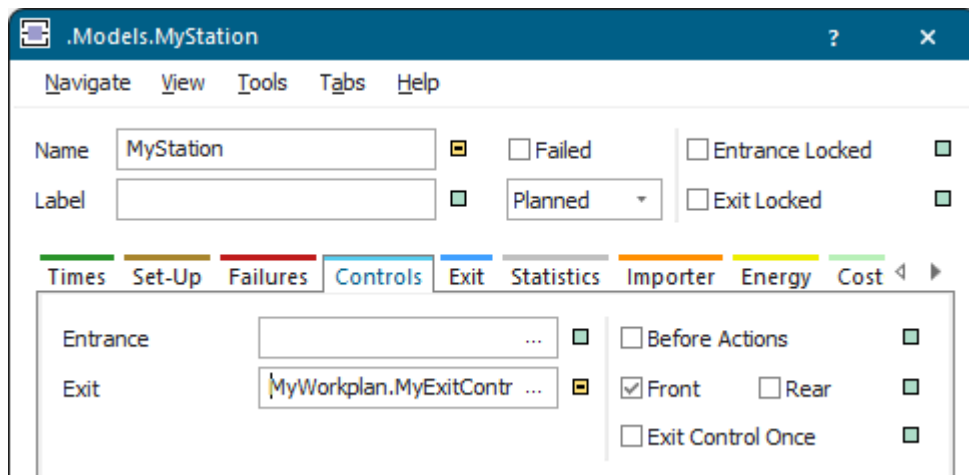


Back to [Create the Processing Stations](#)

Specify the Name of the Exit Control in the Class of the Processing Stations

As the station has to find the next station in the sequence of operations after each processing step and move the part on to this station, we enter the name of an **Exit Control**.

Program the Exit Control shows how to accomplish this.



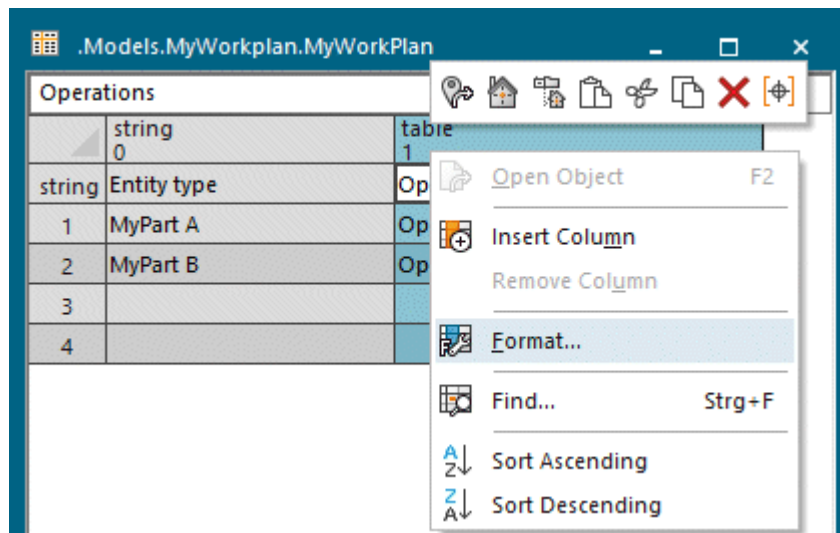
Back to [Create the Processing Stations](#)

Create the Work Plan

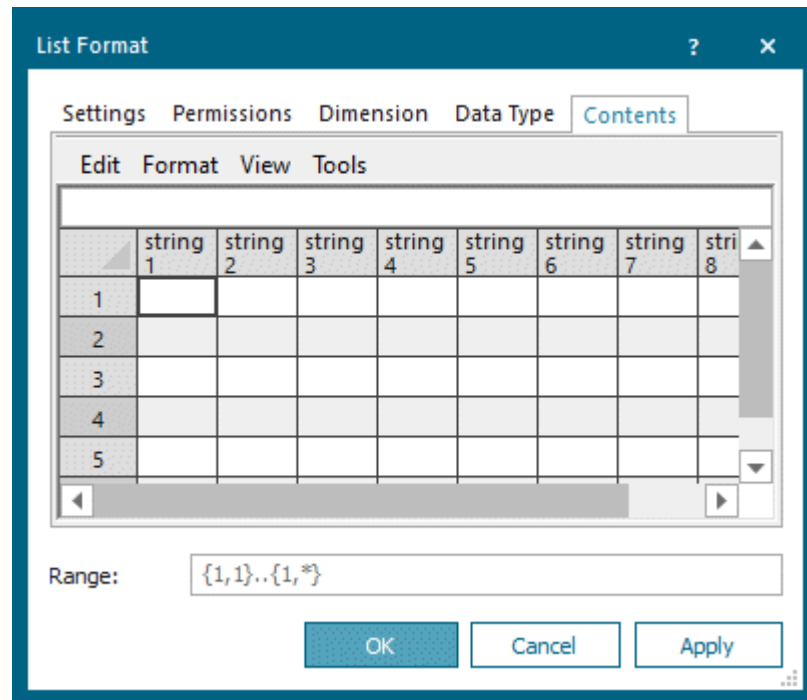
In the last step, we create the work plan.

To create the work plan:

- Insert a *DataTable*  into the *Frame*. Activate the column index and the row index.
- Assign the data type **Table** to the column which contains the subtable into which we enter information about the operations.
- Format this column so that all subtables have the same format, compare [Create Lists within Lists and Tables](#). To do so, click the right mouse button on the column header **Operations** and select **Format**.



Scroll all the way to the right to the tab **Contents** in the dialog **List Format**. Activate the column index and the row index of the subtables in the **Format** menu.



Assign the data type **Object** to the column of the column index. Assign the data type **Time** to the next two columns.

- After we assigned the correct format to the subtables, we can then enter the contents:
 - Once we enter an identifier, *Plant Simulation* creates the subtable in this cell. We typed in Operations for MyPart A and Operations for MyPart B.

	string 0	table 1
string	Entity type	Operations
1	MyPart A	Operations for MyPart A
2	MyPart B	Operations for MyPart B
3		
4		

- To open the subtable for the respective work plan, double-click the respective cell below Operations.
- Enter the names of the operations/stations, their set-up times and the processing times of the parts.

Operations MyPart A				Operations MyPart B			
.Models.MyWorkplan.MyWorkPlan[1,1]				.Models.MyWorkplan.MyWorkPlan[1,2]			
	object 0	time 1	time 2		object 0	time 1	time 2
string	Operation	Setup time	Processing time	string	Operation	Setup time	Processing time
1	Milling	10.0000	1:00.0000	1	Milling	10.0000	1:00.0000
2	Drilling_A	20.0000	2:00.0000	2	Drilling_B	20.0000	2:00.0000
3	Packing	10.0000	3:00.0000	3	Packing	10.0000	3:00.0000
4				4			

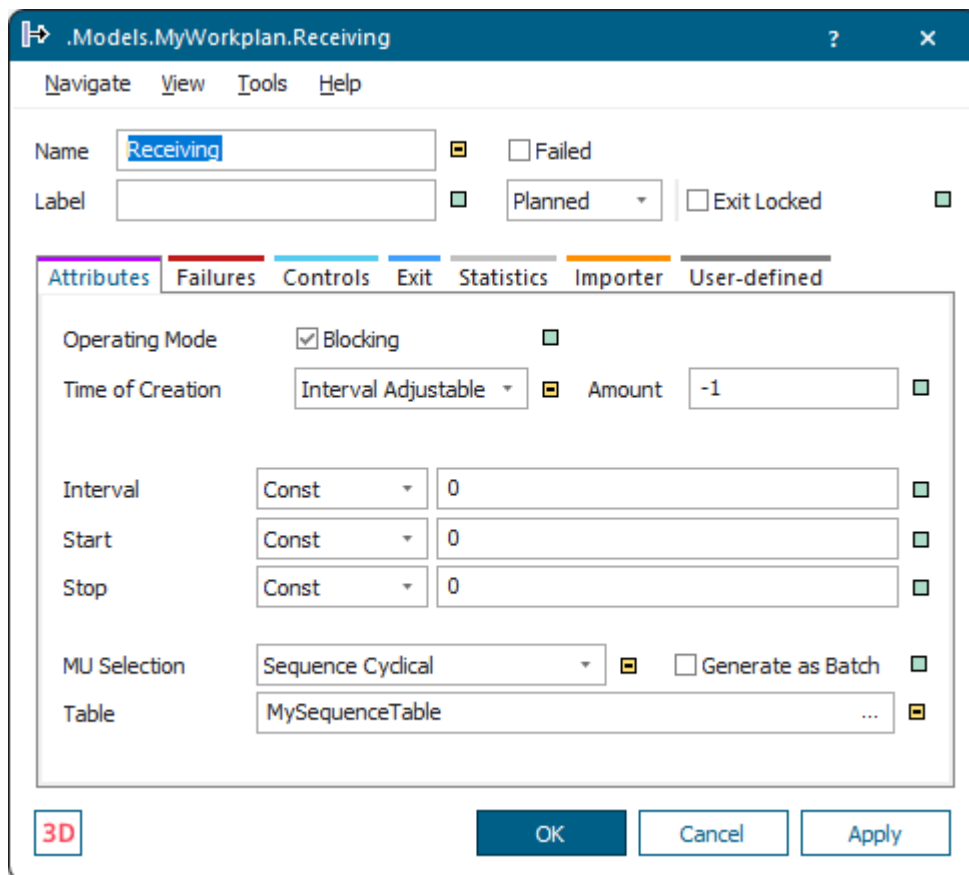
Back to [Produce and Process Parts with a Work Plan](#)

Produce the Parts with a Source Using a Sequence Table

We produce the parts with a source using a sequence table.

To produce the parts in the sequence, which we specify in a sequence table:

- Insert a Source  into the *Frame*. Select **MU Selection** > [Sequence Cyclical](#).



.Models.MyWorkplan.Receiving

Name: ☐ Failed
 Label: ☐ Planned ☐ Exit Locked

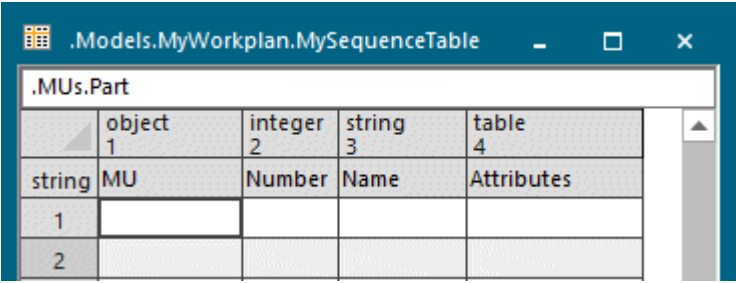
Attributes | Failures | Controls | Exit | Statistics | Importer | User-defined

Operating Mode: ☒ Blocking ☐
 Time of Creation: ☐ Amount: ☐

Interval: ☐
 Start: ☐
 Stop: ☐

MU Selection: ☐ ☐ Generate as Batch ☐
 Table: ☐

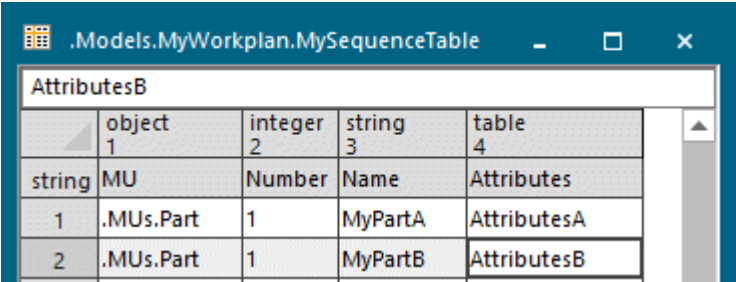
- Insert a *DataTable*  into the *Frame*. Drag this table over the text box **Table. Plant Simulation** automatically assigns the columns on the table the correct data types and the correct column headers.



	object 1	integer 2	string 3	table 4
string	MU	Number	Name	Attributes
1				
2				

- Open the table and enter the type of part you want to produce, how many of them, their name and the name of the subtable containing the attributes of the part.

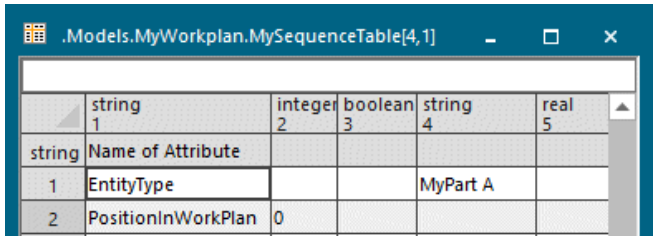
In our example the *Source* is to produce 1 part of the *MU* class *Part* each with the name *MyPartA* and *MyPartB*.



	object 1	integer 2	string 3	table 4
string	MU	Number	Name	Attributes
1	.MUs.Part	1	MyPartA	AttributesA
2	.MUs.Part	1	MyPartB	AttributesB

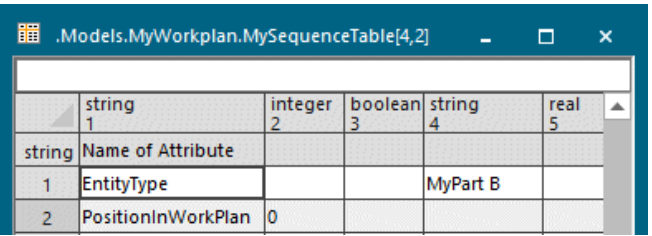
- To create a subtable for the *user-defined attributes* of the parts, enter an identifier into the cells below **Attributes**. To open this subtable, double-click the cell. Here we enter the names and the values of two *user-defined attributes*: One sets the name of the part, the *PartType* is either *MyPartA* or *MyPartB*. The other sets the position of the part in the work plan (*PositionInWorkPlan*). In our case the sequence of operations always starts at position 0, which is the *Source*. The counter, which we programmed in the **Exit Control**, then steps through the stations: *Milling* is position 1, *Drilling_A* or *Drilling_B* are position 2 and *Packing* is position 3.

User-defined Attributes for MyPart A



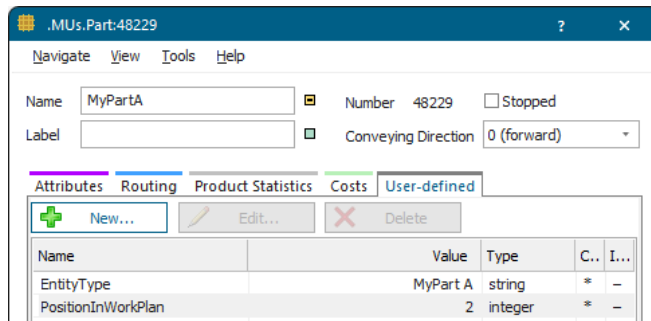
	string 1	integer 2	boolean 3	string 4	real 5
string	Name of Attribute				
1	EntityType			MyPart A	
2	PositionInWorkPlan	0			

User-defined Attributes for MyPart B



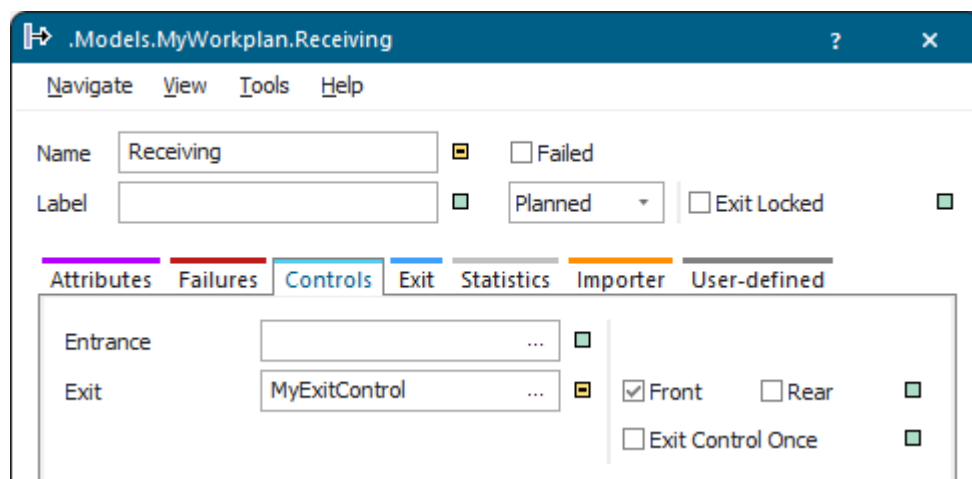
	string 1	integer 2	boolean 3	string 4	real 5
string	Name of Attribute				
1	EntityType			MyPart B	
2	PositionInWorkPlan	0			

During the simulation run *Plant Simulation* enters these *user-defined attributes* into the created *Parts*. The *Part* with the number **48229**, for example, is of type **MyPart A** and is located on station 2, *DrillingA*, according to the position, which we defined in the work plan as operation.



	object 0	time 1	time 2
string	Operation	Setup time	Processing time
1	Milling	10.0000	1:00.0000
2	Drilling_A	20.0000	2:00.0000
3	Packing	10.0000	3:00.0000
4			

- Finally, enter the name of the **Exit Control**, which finds the processing station and moves the part onto it.



Back to Produce and Process Parts with a Work Plan

Program the Exit Control

We want all *stations* and the *Source* to access the source code for finding the next station in the sequence of operations after each processing step and for moving the part on to this station. We do this by programming an **Exit Control**.

If just the stations were going to use this control, we could have programmed it in a *user-defined attribute* of data type *method* of the class of the station.

The source code of the **Exit Control** looks like this:

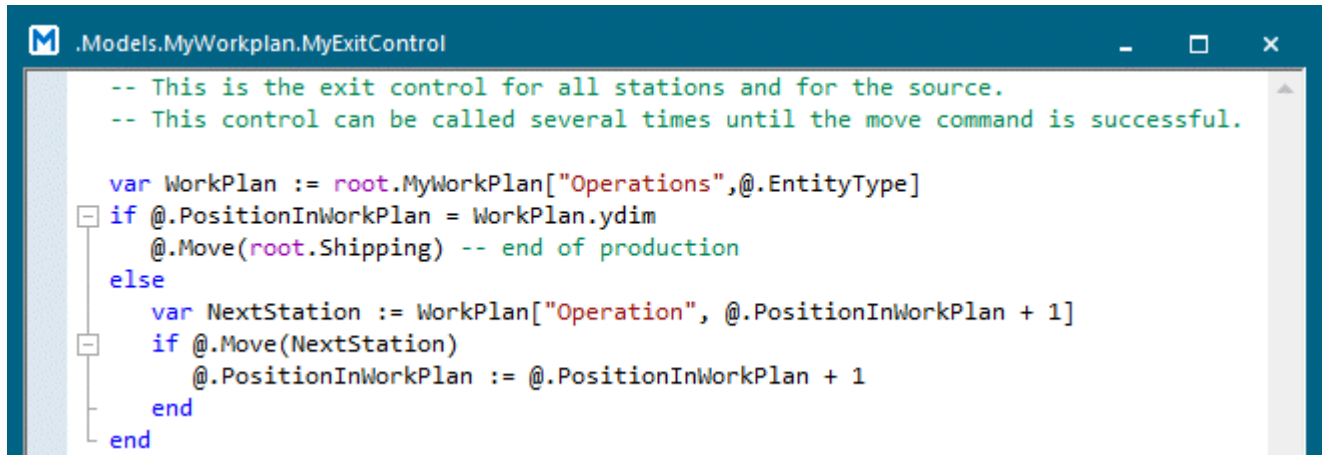
```
// This is the exit control for all stations and for the source.
// This control can be called several times until the move command is
// successful.

var WorkPlan := root.MyWorkPlan["Operations",@.EntityType]
if @.PositionInWorkPlan = WorkPlan.ydim
```

```

    @.Move(root.Shipping) // end of production
else
    var NextStation := WorkPlan["Operation", @.PositionInWorkPlan + 1]
    if @.Move(NextStation)
        @.PositionInWorkPlan := @.PositionInWorkPlan + 1
    end
end
end

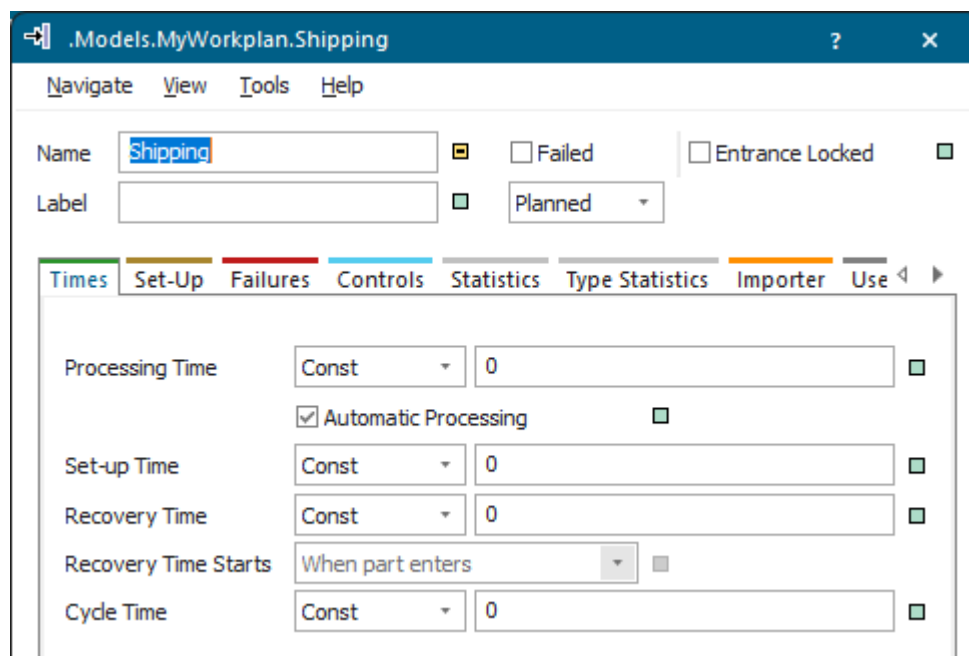
```



Back to [Produce and Process Parts with a Work Plan](#)

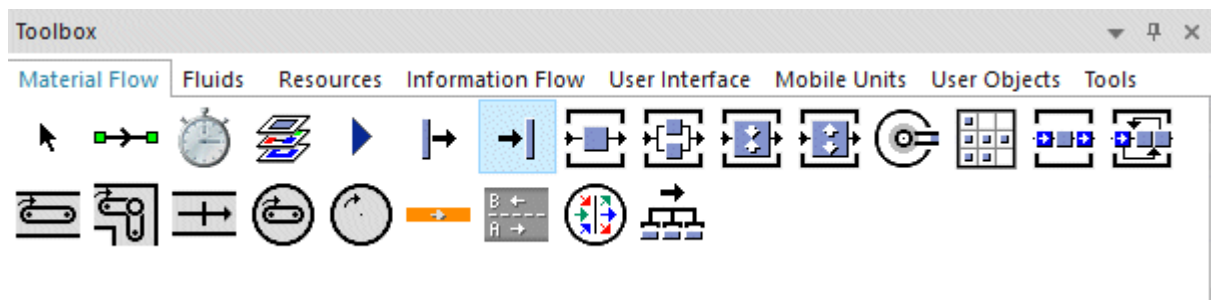
Remove Parts from the Installation with the Drain

To remove the parts and workpieces from the plant after they have been processed, for example to model the *shipping department*, we use the object *Drain*.



The **Drain**  has a single processing place. It removes the *MU* from the installation after setting-up for it and after processing it. The built-in properties of the *Drain* are the same as those of the **Station** . The *Drain* removes the *MU* from the installation after it processed it, instead of moving it on to a succeeding object in the flow of materials, and it collects statistics about the *MU*.

You can insert the *Drain*  into your simulation model from the folder **MaterialFlow** in the *Class Library* or from the toolbar **Material Flow** in the *Toolbox*.



Remember that we create these parts with the object **Source** .

Compare [Specify Times in Object Dialogs](#)

Compare [Check How Many Parts Left the Plant](#)

Compare [Produce Parts with the Source](#)

Back to [Model the Flow of Materials, Basics](#)

Transferring Parts from Station to Station

Transfer Parts from Station to Station

You can transfer parts from station to station in a number of ways.

When transferring parts from station to station in your simulation model, you can either:

- [Use the Standard Transfer Behavior](#), which employs the **push-block** principle.
- [Select an Exit Strategy](#), which distributes the parts to the succeeding stations according to the criteria you set.
- [Distribute Parts Among its Successors with the FlowControl](#).
- Move the part, when an attribute of the processing station changes by using an [observer](#).
- Select an **Exit Strategy** in the [Interface](#) when parts are moving across *Frames*.
- [Load, Unload, and Reload Parts with the TransferStation](#).

Compare the sample models: Click the **Window** ribbon tab, click **Start Page > Getting Started > Example Models > Small Examples**. Then, select the respective **Category**, the **Topic**, and the **Example** in the dialog **Examples Collection**, and click **Open Model**.

Compare [How Plant Simulation Moves Parts On](#)

Back to [Model the Flow of Materials, Basics](#)

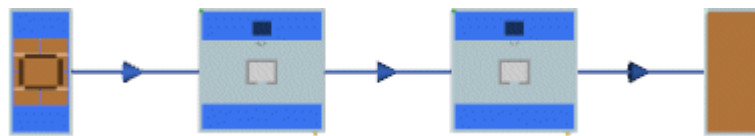
How Plant Simulation Moves Parts On

Plant Simulation moves the parts to be processed through the plant depending on the type of material flow object.

Plant Simulation moves *MUs* on as follows:

- **Moving from point-oriented to point-oriented object**

When moving from a point-oriented object to another point-oriented object, the *MU* moves on completely and instantaneously from booking point to booking point.



- **Moving from point-oriented object to length-oriented object**

When moving from a point-oriented object, such as a *Source* or a *Station*, to a *Conveyor*, the front of the *MU* moves to the start of the *Conveyor*. When *Station* and *Conveyor* are not placed directly side by side, the *MU* seems to hang in thin air, compare [Position](#) of the *MU*.

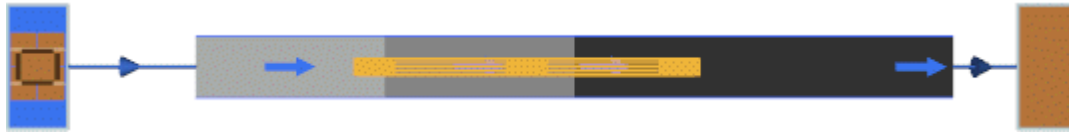


If the objects are placed directly side by side you will not notice that as the *MU* still seems to be located on the predecessor.



- **Moving from length-oriented object to length-oriented object**

When moving from *Conveyor* to *Conveyor*, the *MU* moves **continually** on, i.e., only its front moves on to the successor, while the remainder follows with the [Speed](#) you specified. When the transport speeds differ, *Plant Simulation* uses the speed of the *Conveyor*, on which the [Booking Point Length](#) of the *MU* is located. In the example below our *MU* is located on three *Conveyors* as it is very long.



- **Moving from length-oriented object to point-oriented object**

When moving onto a point-oriented object, such as a *Station*, the *MU* always moves completely and instantaneously, i.e., it is then located on that object as a whole, not only its front section.



Back to [Model the Flow of Materials, Basics](#)

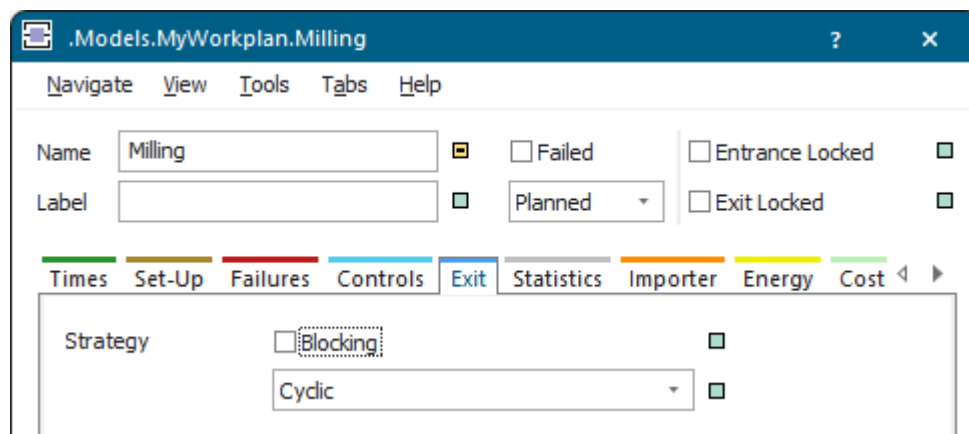
Use the Standard Transfer Behavior

By default *Plant Simulation* uses the **push-block principle** then transferring parts from station to station, either along the connections you established with the *Connector* or by programming a *Method*.

An object that has processed the part, which is ready to be transferred, actively attempts to move that part on to its successor (**push**). If the successor cannot receive the part at the moment, the **block principle** will be activated, guaranteeing that the object ready to move the part is re-activated as soon the successor is ready to receive the part.

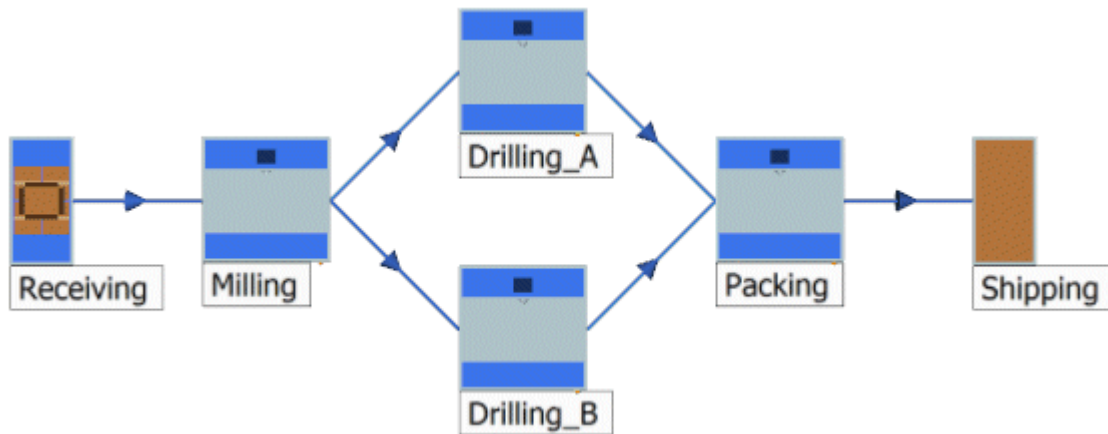
The push-block principle ensures that parts are transferred using the built-in functionality of the objects and that the stream of events is not interrupted by failures or pauses.

The standard transfer strategy is cyclic, non-blocking, meaning that the current object moves the *MU* to the first non-blocked succeeding object in the flow of materials. Once the object reaches the end of the list of successors, it continues the search at the beginning of the list and terminates the search when it reaches the successor from the previous search.



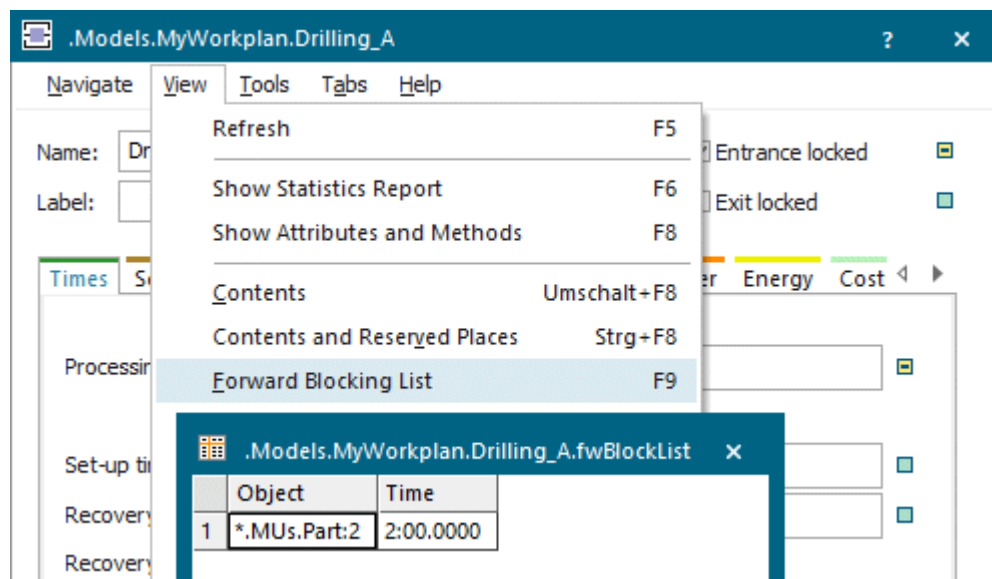
Suppose *Milling* is the first station in the sequence of stations in your simulation model, *Drilling_A* is the second station, *Part:1* is the first mobile object that entered the flow of materials, *Part:2* is the second mobile object.

Suppose *Part:1* is located on *Drilling_A*, and *Part:2* is located on *Milling* and wants to move to *Drilling_A*.



Once *Part:2* is completely processed by *Milling*, *Part:2* notifies *Drilling_A* that it intends to move to it.

- When *Drilling_A* can receive *Part:2*, it notifies *Part:2*, and *Part:2* moves from *Milling* to *Drilling_A*.
- When *Drilling_A* cannot receive *Part:2* because it is busy, its entrance is locked, etc., *Part:2* enters itself into the **forward blocking list** of *Drilling_A* and of all other stations it intends to move to.



- When *Part:1* exits *Drilling_A*, *Drilling_A* schedules an **Out** event for all entries, i.e., *MUs*, in the **Forward Blocking List**, including *Part:2*. Afterwards *Drilling_A* deletes all entries from its **Forward Blocking List**.
- Provided *Drilling_A* can receive *Part:2*, *Part:2* deletes all references to itself from all blocking lists of all stations it entered itself into, and then moves to *Drilling_A*.

The material flow objects *Station*, *ParallelStation*, *AssemblyStation*, *DismantleStation*, *Conveyor*, *Sorter*, *PlaceBuffer* and *Buffer* handle the *MUs* that transfer onto them:

Procedure

1. For these times, which you define:

- The **Processing Time** is the time, which the *MU* remains on the object to be processed. It is the interval between setting-up for the present type of *MU* and the point in time when the material flow object moves it on to its successor.
- The **Set-up Time** is the time it takes to set-up the object for processing a different type of *MU*. An identical name designates that *MUs* are of the same type and that the object does not have to set up for this *MU*.
- The **Recovery Time** is the time which is required to set a station into a defined state before it can start processing the next part. The recovery time is, for example, useful to model materials handling equipment, such as a robot, which requires a certain time to insert work pieces into or remove them from processing stations. You can also use it to model times during which a hot part cools off.
- The **Cycle Time** is the time that passes between the points in time where parts can enter the object. The cycle time is, for example, useful to model chain conveyors with a fixed chain interval that only transport materials when a free hook is available.

The screenshot shows the configuration window for a material flow object named 'Milling'. The 'Times' tab is selected, displaying various time-related parameters. The 'Processing Time' is set to a formula 'ProcessingTimeInFormula'. The 'Set-up Time' is also a formula, 'root.MyWorkPlan["Operations",@.EntityT]', with 'Automatic Processing' checked. 'Recovery Time' is set to a constant '0'. 'Recovery Time Starts' is set to 'When part enters'. 'Cycle Time' is set to a constant '0'. There are checkboxes for 'Failed', 'Entrance Locked', and 'Exit Locked'.

Note

You do not have to define all of the times listed above for the object.

2. For **failures** which you define and activate.

The material flow objects then transfer the *MUs* to the next object along the material flow connections using the **Exit Strategy** you select.

Compare [Define Processing Times](#)

Compare [Model Failures](#)

Compare [Select the Set-Up Time](#)

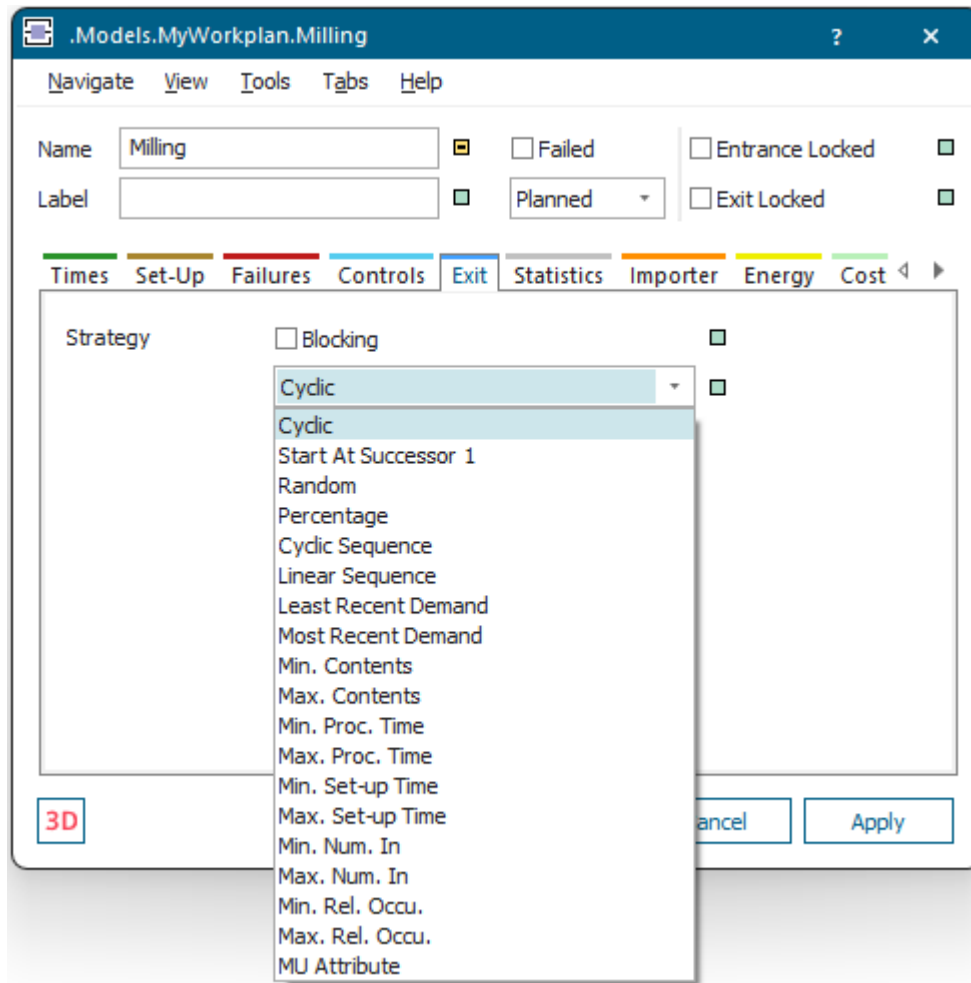
Back to [Transfer Parts from Station to Station](#)

Selecting an Exit Strategy

Select an Exit Strategy

If you do not want to transfer the *MUs* using the standard transfer behavior, select an **Exit Strategy**.

To do so, click the [Tab Exit](#) and select another [Strategy](#). As a rule you will use an **Exit Strategy**, which distributes the flow of materials, when the sequence of operations does not uniquely designate the next object that processes the *MU*. Your objective will be to make a choice that is close to optimal in some respect, say cost or time, but not in others.

**Note**

For length-oriented objects *Plant Simulation* only considers the **Exit Strategy** when the part is moving forward.

Note

Click **Apply**, to apply your new settings, and to display additional dialog items, which the selected exit strategy requires.

- To move the part cyclically to the next successor in line, select **Cyclic**.
- To cyclically move the part to the successor according to the sequence of successors, which you typed into the list, select **Cyclic Sequence**.

You can also set the number of the succeeding object with the attribute **ExitStrategySequence**.

- To move the part to the successor that has been **Waiting** the longest for a *MU*, select **Least Recent Demand**.

- To move the part to the successor in a linear fashion, according to the sequence of successors, which you typed into the list, select **Linear Sequence**.

You can also set the number of the succeeding object with the attribute **ExitStrategySequence**.

- To move the part to the successor that contains the greatest number of *MUs*, select **Maximum Contents**.
- To move the part to the successor that received the most *MUs*, select **Maximum Number In**.
- To move the part to the successor with the longest processing time, select **Maximum Processing Time**.
- To move the part to the successor with the highest relative occupancy, select **Maximum Relative Occupation**.
- To move the part to the successor with the longest set-up time, select **Maximum Set-up Time**.
- To move the part to the successor that contains the smallest number of *MUs*, select **Minimum Contents**.
- To move the part to the successor that received the smallest number of *MUs*, select **Minimum Number In**.
- To move the part to the successor with the shortest processing time, select **Minimum Processing Time**.
- To move the part to the successor with the lowest relative occupancy, select **Minimum Relative Occupation**.
- To move the part to the successor with the shortest set-up time, select **Minimum Set-up Time**.
- To move the part to the successor that has been **Waiting** the least amount of time for a *MU*, select **Most Recent Demand**.
- To move the part to the successor according to an attribute of the *MU*, select **MU Attribute**.

You can also set the names of the user-defined or the built-in attributes, their values, and the numbers of the successors with the attribute **ExitStrategyMUAttributeList**.

- To move the part to the successors according to a percentage distribution, select **Percentage**.

You can also set the with the attribute **ExitStrategyPercentageValues**.

- To move the part to the successors in a random fashion, select **Random**.

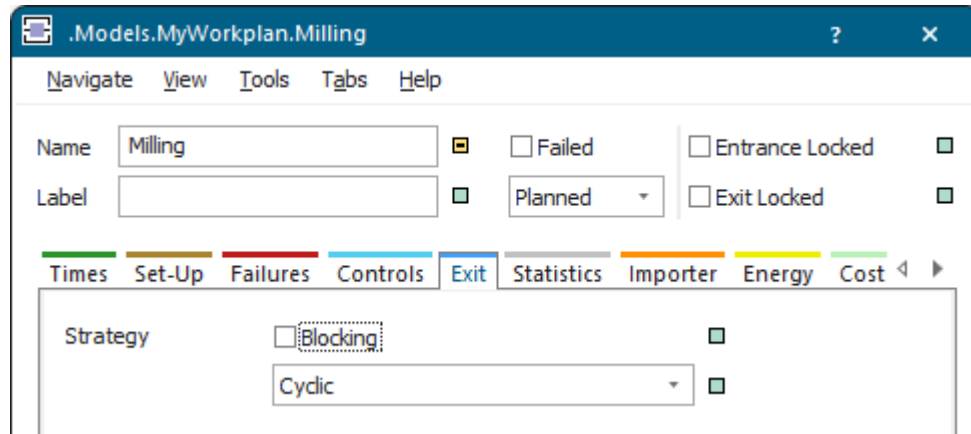
You can also set the with the attribute **ExitStrategyPercentageValues**.

- To always move the part to successor with the number 1, select **Start at Successor 1**.

Back to [Transfer Parts from Station to Station](#)

Cyclic [exit strategy, example]

To move the part cyclically to the next successor in line, select the **Exit Strategy > Cyclic** from the drop-down list.



- Select **Blocking** ☒ **Blocking**  to move the *MU* to the object, which in sequence of successors immediately follows on the object on to which a *MU* has last transferred.
- Clear **Blocking** ☐ **Blocking**  to move the *MU* to the next object in the sequence of successors, which can receive a part.

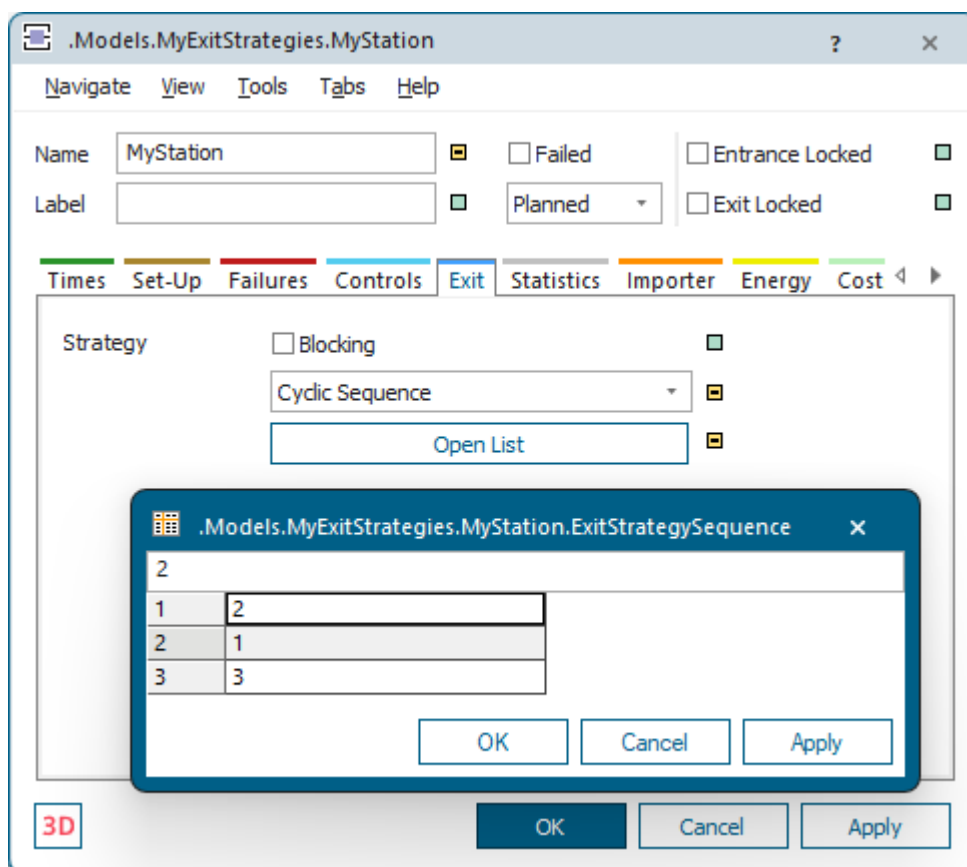
Once the object reaches the end of the sequence of successors, it continues the search at the beginning of the sequence and terminates the search when it reaches the successor from the previous search.

See also [Cyclic \[described\]](#)

Back to [Select an Exit Strategy](#)

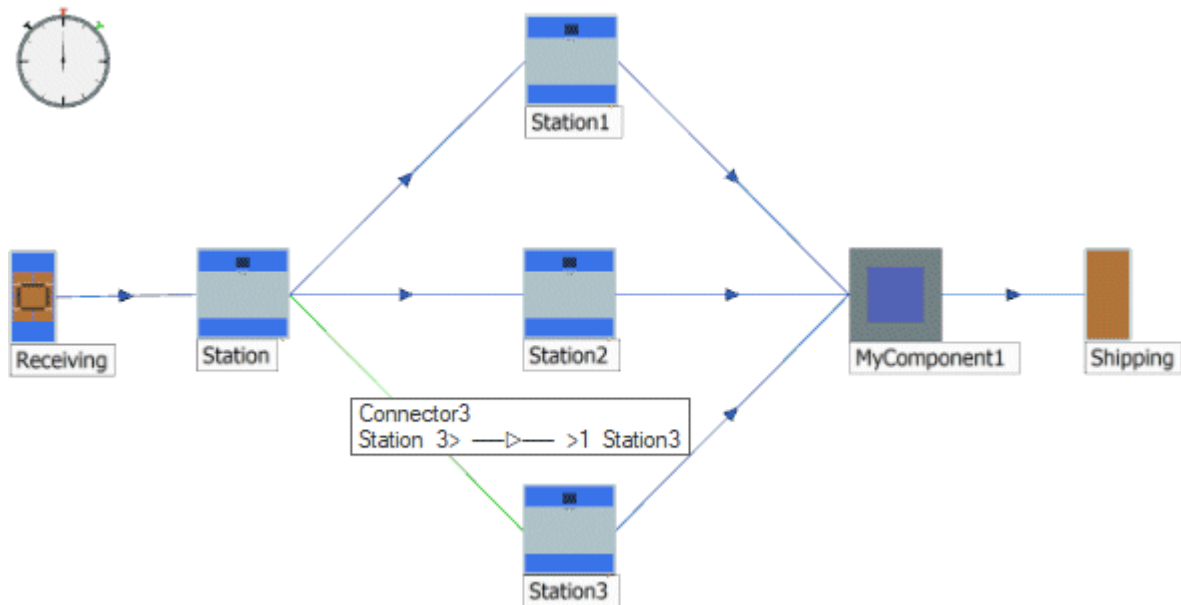
Cyclic Sequence [example]

To cyclically move the part to the material flow object's successor according to the sequence of successors repeatedly, select the **Exit Strategy > Cyclic Sequence** from the drop-down list.

**Note**

Click **Apply**, to apply your settings, and to display the button **Open List**.

Click **Open List** and type in the number of the succeeding object into the respective cell of the list. When you type in 2 into the cell in row 1, the object first moves the *MU* to the successor with the number 2.



The successor to which the object moves the *MU* is the next one in line that can receive it in the sequence, which you typed into the list. When the object arrives at the last entry in the list (last object in the sequence), it returns to the first entry and processes the list again.

The same successor object may be contained several times, one after the other, in the list. Then the object attempts to move the part to this same successor as often as it is contained in the list.

- Select **Blocking** ☒ **Blocking** to only move the *MU*, when the designated successor is ready to receive it.
- Clear **Blocking** ☐ **Blocking** to attempt to move the part to the respective successor, beginning with the active entry of the list.

When the object arrives at the last entry in the list, i.e., the last object in the sequence, it returns to the first entry and processes the list again.

See also [Cyclic Sequence \[described\]](#)

Back to [Select an Exit Strategy](#)

Least Recent Demand [example]

To move the part to the successor that has been **Waiting** the longest for a *MU*, select the **Exit Strategy > Least Recent Command** from the drop-down list.

This means that the successor receives the next *MU* onto which no part moved for the longest time.

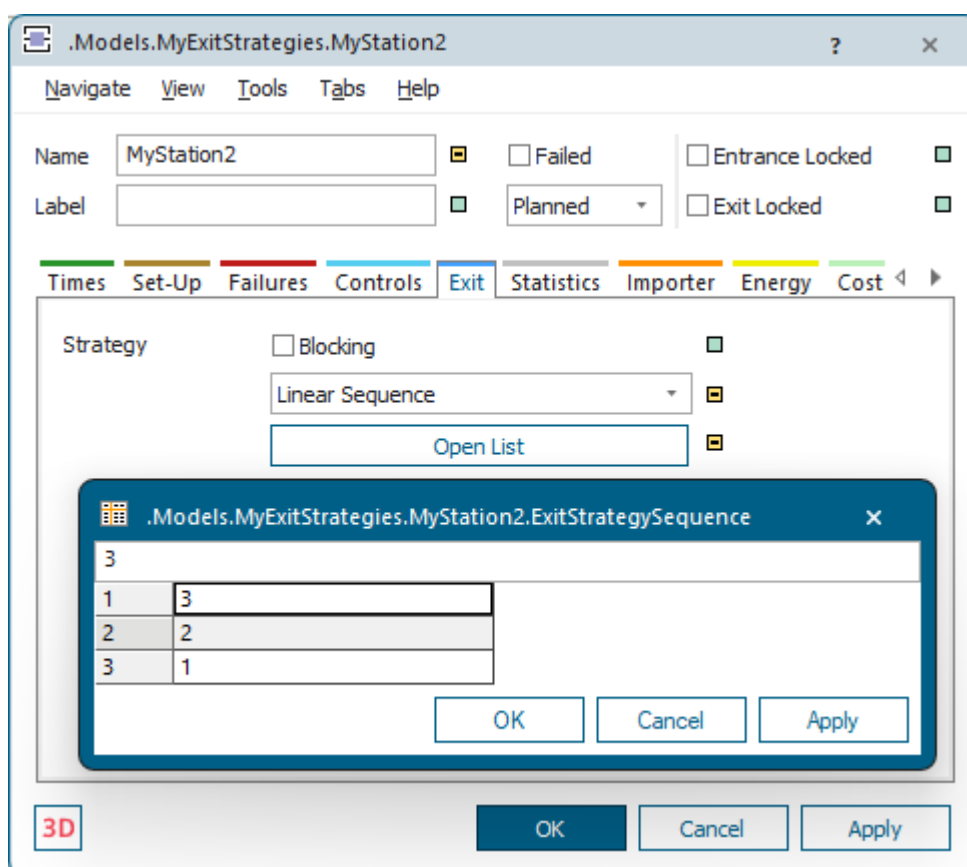
- Select **Blocking** ☒ **Blocking** to move the *MU* to the next successor.
- Clear **Blocking** ☐ **Blocking** to move the *MU* to the next successor, which can receive the *MU*.

See also [Waiting \[state, material flow objects\]](#) [Least Recent Demand \[described\]](#)

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Linear Sequence [example]

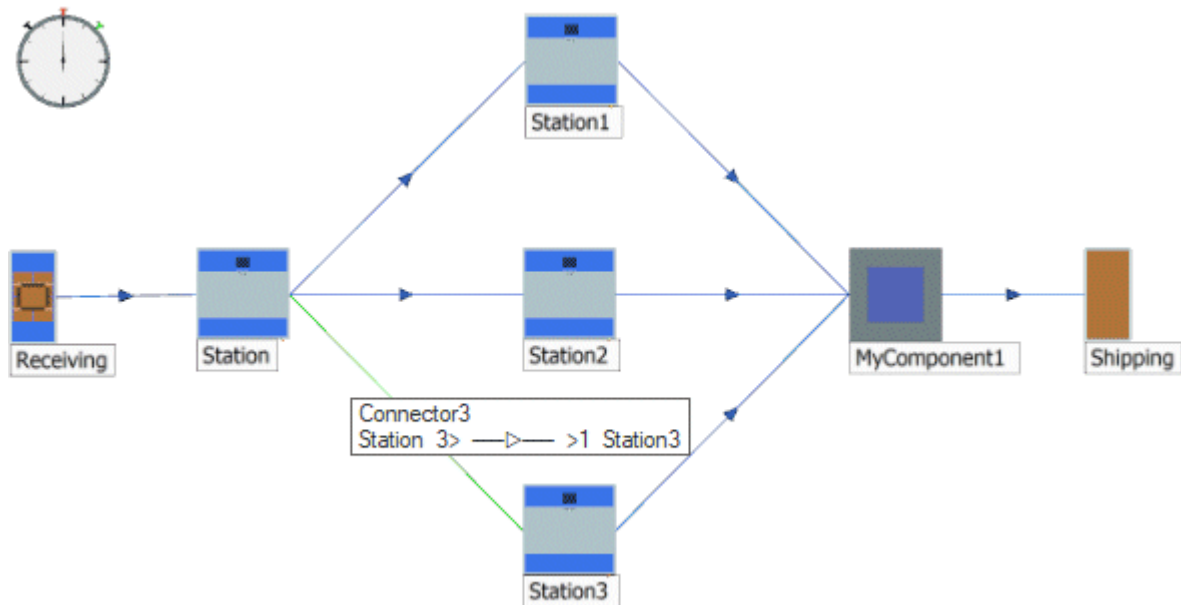
To move the part to the material flow object's successor in a linear fashion, one time only, according to the sequence of successors, select the **Exit Strategy** > **Linear Sequence** from the drop-down list.



Note

Click **Apply**, to apply your settings, and to display the button **Open List**.

Click **Open List** and type in the number of the succeeding object into the corresponding cell of the list. When you type in 3 into the cell in row 1, the object first moves the *MU* to the successor with the number 3.



The successor to which the object moves the *MU* is the first one in line that can receive it in the sequence, which you typed into the list.

- Select **Blocking** ☒ Blocking to only move the *MU* to the first successor in the list of successors.
- Clear **Blocking** ☐ Blocking to move the *MU* to the first available successor in the list of successors.

See also [Linear Sequence \[described\]](#)

Back to [Select an Exit Strategy](#)

Maximum Contents

To move the part to the successor that contains the highest number of *MUs*, select the **Exit Strategy > Maximum Contents** from the drop-down list.

Note

This only works correctly, when resource statistics of the successor or of the predecessor is active.



- Select **Blocking** ☒ Blocking to move the *MU* to the next successor.
- Clear **Blocking** ☐ Blocking to move the *MU* to the next successor, which can receive the *MU*.

See also [Max. Contents](#)

Back to [Select an Exit Strategy](#)

Maximum Number In

To move the part to the successor that received the most *MUs*, select the **Exit Strategy > Max. Num. In** from the drop-down list.

Note

This only works correctly, when resource statistics of the successor or of the predecessor is active.



- Select **Blocking** ☒ Blocking  to move the *MU* to the next successor.
- Clear **Blocking** ☐ Blocking  to move the *MU* to the next successor, which can receive the *MU*.

See also [Max. Num. In](#)

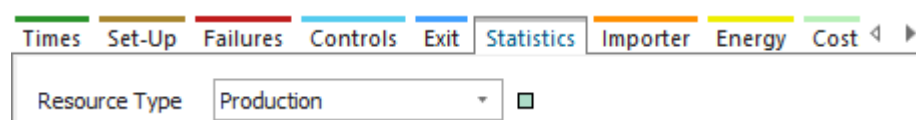
Back to [Select an Exit Strategy](#)

Maximum Processing Time

To move the part to the successor with the longest processing time, select the **Exit Strategy > Max. Proc. Time** from the drop-down list.

Note

This only works correctly, when resource statistics of the successor or of the predecessor is active.



- Select **Blocking** ☒ Blocking  to move the *MU* to the next successor.
- Clear **Blocking** ☐ Blocking  to move the *MU* to the next successor, which can receive the *MU*.

See also [Max. Proc. Time](#)

Back to [Select an Exit Strategy](#)

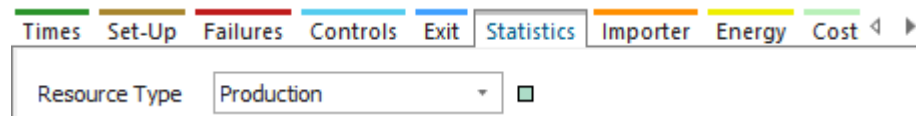
Maximum Relative Occupation

To move the part to the successor with the highest relative occupancy, select the **Exit Strategy > Max. Rel. Occu.** from the drop-down list.

To move the part to the successor with the highest relative occupancy, select **Max. Rel. Occu.** from the drop-down list.

Note

This only works correctly, when resource statistics of the successor or of the predecessor is active.



- Select **Blocking** ☒ **Blocking**  to move the *MU* to the next successor.
- Clear **Blocking** ☐ **Blocking**  to move the *MU* to the next successor, which can receive the *MU*.

See also [Max. Rel. Occu.](#)

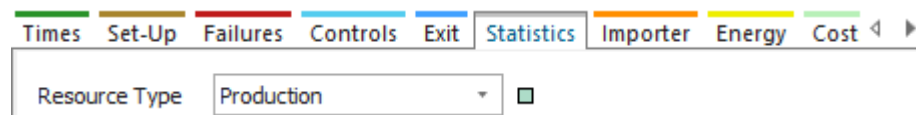
Back to [Select an Exit Strategy](#)

Maximum Set-up Time

To move the part to the successor with the longest set-up time, select the **Exit Strategy > Max. Set-up Time** from the drop-down list.

Note

This only works correctly, when resource statistics of the successor or of the predecessor is active.



- Select **Blocking** ☒ **Blocking**  to move the *MU* to the next successor.
- Clear **Blocking** ☐ **Blocking**  to move the *MU* to the next successor, which can receive the *MU*.

See also [Max. Set-up Time](#)

Back to [Select an Exit Strategy](#)

Minimum Contents

To move the part to the successor that contains the least number of *MUs*, select the **Exit Strategy > Min. Contents** from the drop-down list.

Note

This only works correctly, when resource statistics of the successor or of the predecessor is active.



- Select **Blocking** ☒ **Blocking** to move the *MU* to the next successor.
- Clear **Blocking** ☐ **Blocking** to move the *MU* to the next successor, which can receive the *MU*.

See also [Min. Contents](#)

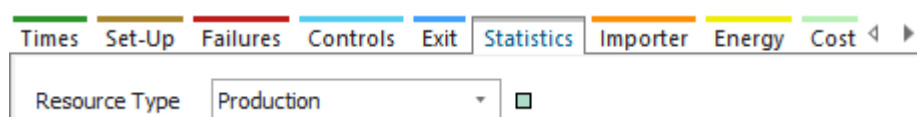
Back to [Select an Exit Strategy](#)

Minimum Number In

To move the part to the successor that received the least number of *MUs*, select the **Exit Strategy > Min. Num. In** from the drop-down list.

Note

This only works correctly, when resource statistics of the successor or of the predecessor is active.



- Select **Blocking** ☒ **Blocking** to move the *MU* to the next successor.
- Clear **Blocking** ☐ **Blocking** to move the *MU* to the next successor, which can receive the *MU*.

See also [Min. Num. In](#)

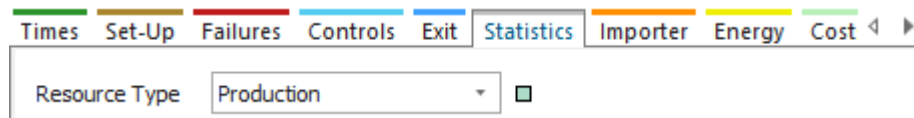
Back to [Select an Exit Strategy](#)

Minimum Processing Time

To move the part to the successor with the shortest processing time, select the **Exit Strategy > Min. Proc. Time** from the drop-down list.

Note

This only works correctly, when resource statistics of the successor or of the predecessor is active.



- Select **Blocking** ☒ **Blocking** to move the *MU* to the next successor.
- Clear **Blocking** ☐ **Blocking** to move the *MU* to the next successor, which can receive the *MU*.

See also [Min. Proc. Time](#)

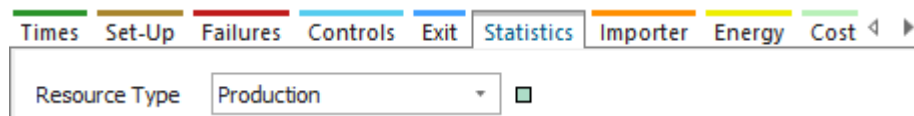
Back to [Select an Exit Strategy](#)

Minimum Relative Occupation

To move the part to the successor with the lowest relative occupancy, select the **Exit Strategy > Min. Rel. Occu.** from the drop-down list.

Note

This only works correctly, when resource statistics of the successor or of the predecessor is active.



- Select **Blocking** ☒ **Blocking** to move the *MU* to the next successor.
- Clear **Blocking** ☐ **Blocking** to move the *MU* to the next successor, which can receive the *MU*.

See also [Min. Rel. Occu.](#)

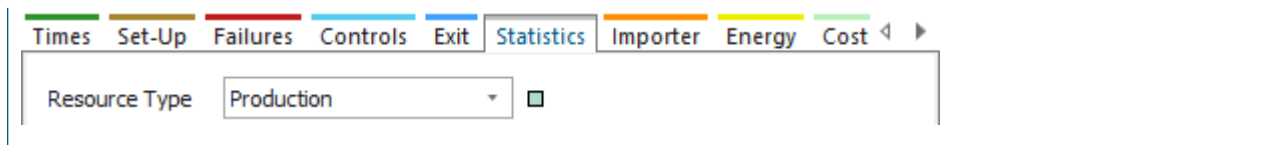
Back to [Select an Exit Strategy](#)

Minimum Set-up Time

To move the part to the successor with the shortest set-up time, select the **Exit Strategy > Min. Set-up Time** from the drop-down list.

Note

This only works correctly, when resource statistics of the successor or of the predecessor is active.



- Select **Blocking** ☒ **Blocking**  to move the *MU* to the next successor.
- Clear **Blocking** ☐ **Blocking**  to move the *MU* to the next successor, which can receive the *MU*.

See also [Min. Set-up Time](#)

Back to [Select an Exit Strategy](#)

Most Recent Demand [example]

To move the part to the successor **Waiting** the least amount of time for a *MU*, select **Exit Strategy > Most Recent Demand** from the drop-down list.

This means that the successor, onto which a *MUs* was moved last, receives the next *MU*.

- Select **Blocking** ☒ **Blocking**  to move the *MU* to the next successor.
- Clear **Blocking** ☐ **Blocking**  to move the *MU* to the next successor, which can receive the *MU*.

See also [Waiting \[state, material flow objects\]](#) [Most Recent Demand \[described\]](#)

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MU Attribute [example]

To move the part on to a successor according to the value of an attribute of the *MU*, select **Exit Strategy > MU Attribute** from the drop-down list.

Note

Click **Apply** to display the dialog items of this strategy.

- Select **Blocking** to always move the *part* to the designated successor. If that successor is not ready to receive it, the *part* will be blocked.
- Clear **Blocking** to move the *part* on when any of its successors can receive it. In this case enter the desired **numbers** of the successors into the **attribute list** for each **attribute value**.

Name: MyStation
 Label:
 Failed: ☐ Entrance Locked: ☐
 Planned: ☒ Exit Locked: ☐
 Strategy: ☒ Blocking
 MU Attribute:
 Open List
 Default Successor: 1
 Attribute Type: String

Click **Open List** to open a table, into which you can enter the names of the user-defined or the built-in attributes, their values, and the numbers of the successors. The object searches the table from top to bottom until it finds an attribute with the value you entered. It then moves the *MU* on to that successor.

string 1	string 2	integer 3
string Attribute	Value	Successor
1 Color	red	1
2	yellow	2
3	green	3
4 Shape	round	1
5	square	3
6	rectangular	2
7		

OK Cancel

Note

Create the user-defined attributes for the *MUs* on the [Tab User-defined](#).

- **Default Successor**

Enter the number of the **Default Successor**. This is the successor to which the object moves the *MU*, when this *MU* does not have an attribute with the **Value** you entered into the *DataTable*.

To show a message, when a part is to be moved which does not meet any of the conditions in the *DataTable*, enter 0.

- **Attribute Type**

Select the data type of the attribute from this drop-down list, compare [Data Types](#).

See also [MU Attribute \[described\]](#)

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Video on YouTube

<https://youtu.be/PQhEriOzVzU?si=rB6pUI7icKnIr0Yj&t=472>

Percentage [exit strategy, example]

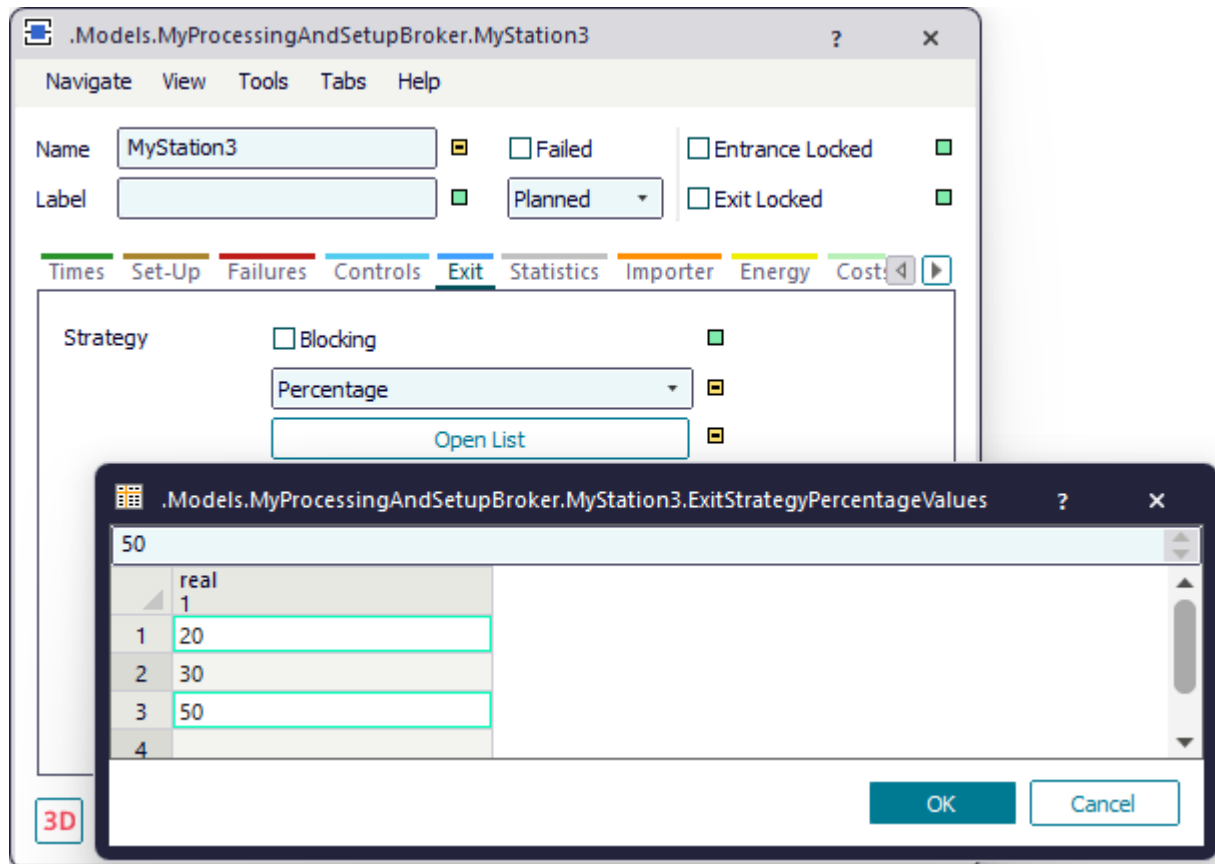
To move the part to the successors according to a percentage distribution, select **Exit Strategy > Percentage** from the drop-down list.

Note

Click **Apply**, to apply your settings, and to display the button **Open List**.

Click **Open List** and type the percentages in the list that opens. The n-th row on the table defines the n-th successor's portion: When you type 20 in row 1, for example, the object moves 20 percent of the *MUs* it received to the successor with the number 1, etc.

The object always moves the *MU* to the successor with the greatest difference between the rated value and the current value.



- Select **Blocking** ☒ **Blocking**  to move the *MU* to the successor with the highest deviation from the nominal percentage.
- Clear **Blocking** ☐ **Blocking**  to move the *MU* to the successor, which can receive the *MU* with the highest deviation from the nominal percentage.

Note

The actual percentages reached may differ from the nominal ones due to a low number of total transfers and to the **Blocking**-state of successors.

Any previous transfers of *MUs* affect the **Percentage** strategy, as they alter the deviation from the nominal transfer frequencies.

Plant Simulation sums up the values you typed into the list of nominal percentages to obtain the value that matches 100 percent. The distribution pattern only depends on the relative size of the values, not on their magnitude. This way you can, for example, either type in [1; 2] or [0.3333...; 0.6666...], the result will be the same.

If the distribution pattern is periodic, then it is so right from the beginning. This means that the distribution pattern at the start of a simulation run does not differ from the distribution pattern at a later point in time. These examples demonstrate, how nicely the values are balanced:

- If you assign the values 1 and 2 to the exits 1 and 2 respectively, *Plant Simulation* generates this sequence of exits:

2 1 2 2 1 2 2 1 2 2 1 2 2 1 2

- If you assign 2 and 3 to the exits 1 and 2, *Plant Simulation* generates this sequence of exits:

2 1 2 1 2 2 1 2 1 2 2 1 2 1 2

- If you assign the values 1, 2, 4 and 8 to the exits 1, 2, 3, and 4, *Plant Simulation* generates this sequence of exits:

4 3 4 2 4 3 4 1 4 3 4 2 4 3 4 4 3 4 2 4 3 4 1 4 3 4 2 4 3 4

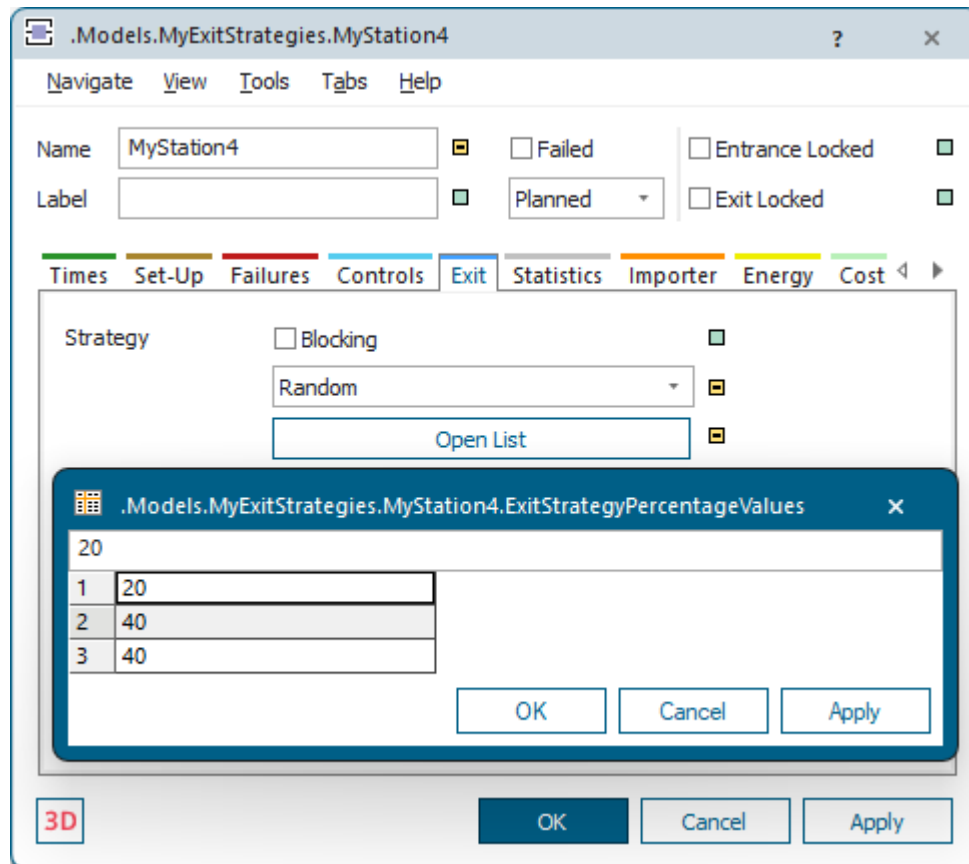
See also [Percentage \[described\]](#)

Back to [Select an Exit Strategy](#)

Random [exit strategy, example]

To move the part to the successors in a random fashion, select the **Exit Strategy > Random** from the drop-down list.

Click **Open List** and type the percentages in the list that opens. The n-th row on the table defines the n-th successor's portion: When you type 20 into row 1, for example, the object moves 20 percent of the *MUs* it received to the successor with the number 1, etc.



- Select **Blocking** ☒ **Blocking** to determine the successor using the built-in random-number generator.
- Clear **Blocking** ☐ **Blocking** to determine the successor using the built-in random-number generator until it either computes a successor, which can receive the *MU* or the object has iterated through all successors.

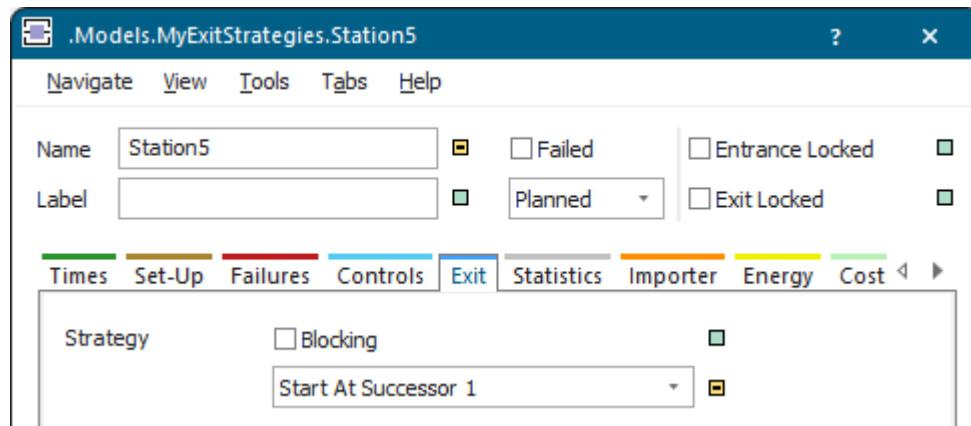
A previous call to the random number generator does affect the **Random** strategy. The order in which the successors are iterated through is not predictable.

See also [Random \[described\]](#)

Back to [Select an Exit Strategy](#)

Start at Successor 1 [example]

To always move the part to the material flow object's successor with the number 1, select the **Exit Strategy > Start At Successor 1** from the drop-down list.



- Select **Blocking** ☒ **Blocking**  to move the *MU* to the first successor in the list of successors.
- Clear **Blocking** ☐ **Blocking**  to move the *MU* to the first successor in the sequence of successors, which can receive the *MU*.

A previous search for a successor does not affect the strategy **Start at successor 1**.

See also [Start at Successor 1 \[described\]](#)

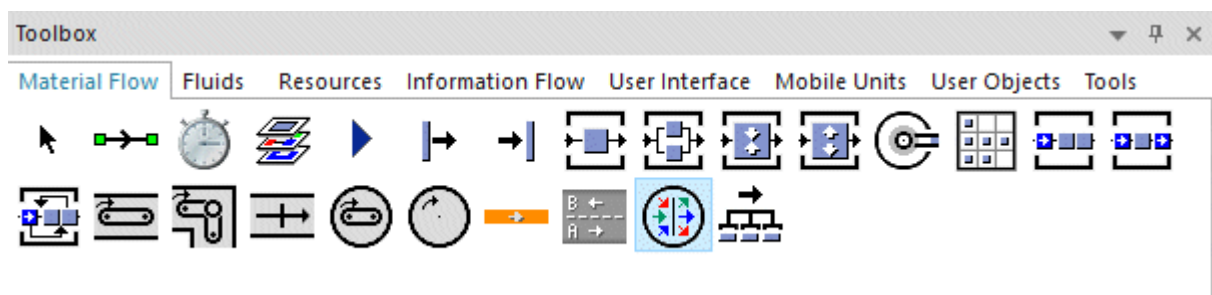
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Distributing Parts with the FlowControl

Distribute Parts Among its Successors with the FlowControl

To model common strategies for splitting-up and for bringing together the flow of materials in your plant, you can use the **FlowControl** . The *FlowControl* does not store and does not process the *MU*, it only distributes them among the objects that succeed it in the sequence of stations in the simulation model. The number of predecessors and successors of the *FlowControl* usually is not limited.

You can insert the *FlowControl* into your simulation model from the folder **Material Flow** in the *Class Library* or from the toolbar **Material Flow** in the *Toolbox*.

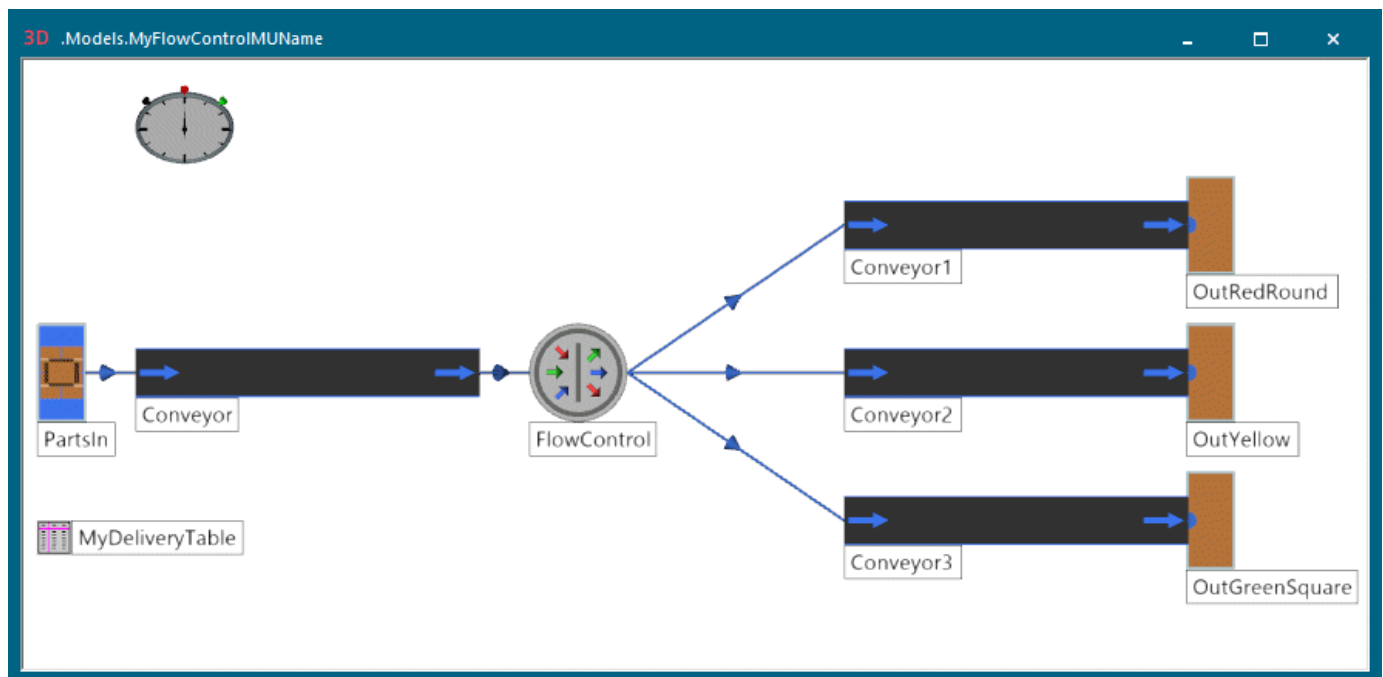


In our example below we show how the *FlowControl* moves the parts on to its successors according to an attribute of the part. If a part has more than one attribute, it suffices if one of the attributes meets the

criteria, which you defined in the attribute list. If a part has several attributes though, and none of these attributes meet the criteria in the list, you can determine what happens to the part by typing in different values for the **default successor** of the *FlowControl*:

- When you type in the number of a successor, the *FlowControl* moves the part so that successor. When you accept the default setting **1**, it moves the part on to the successor object with the number 1.
- When you type in **0**, the *FlowControl* does not move the part on. It remains in front of the *FlowControl*, thus blocking the following parts.
- When you type in **-1**, *Plant Simulation* shows an error message.

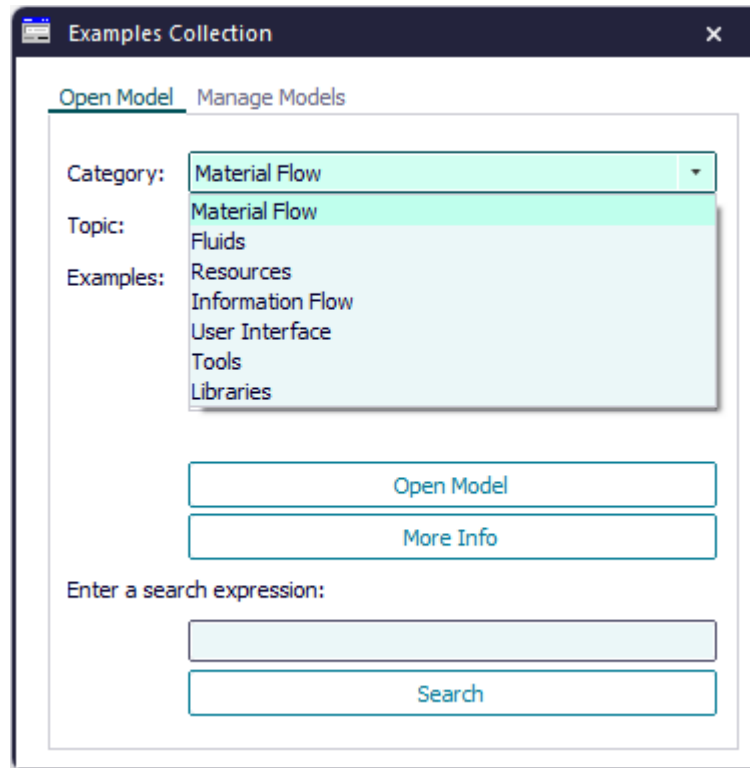
The finished simulation model looks like this:



To create the sample model, you will:

- **Configure the Source that Produces the Parts [FlowControl]**
- **Configure the FlowControl Which Distributes the Parts**

Compare the sample models: Click the **Window** ribbon tab, click **Start Page > Getting Started > Example Models > Small Examples**. Then, select the respective **Category**, the **Topic**, and the **Example** in the dialog **Examples Collection**, and click **Open Model**.

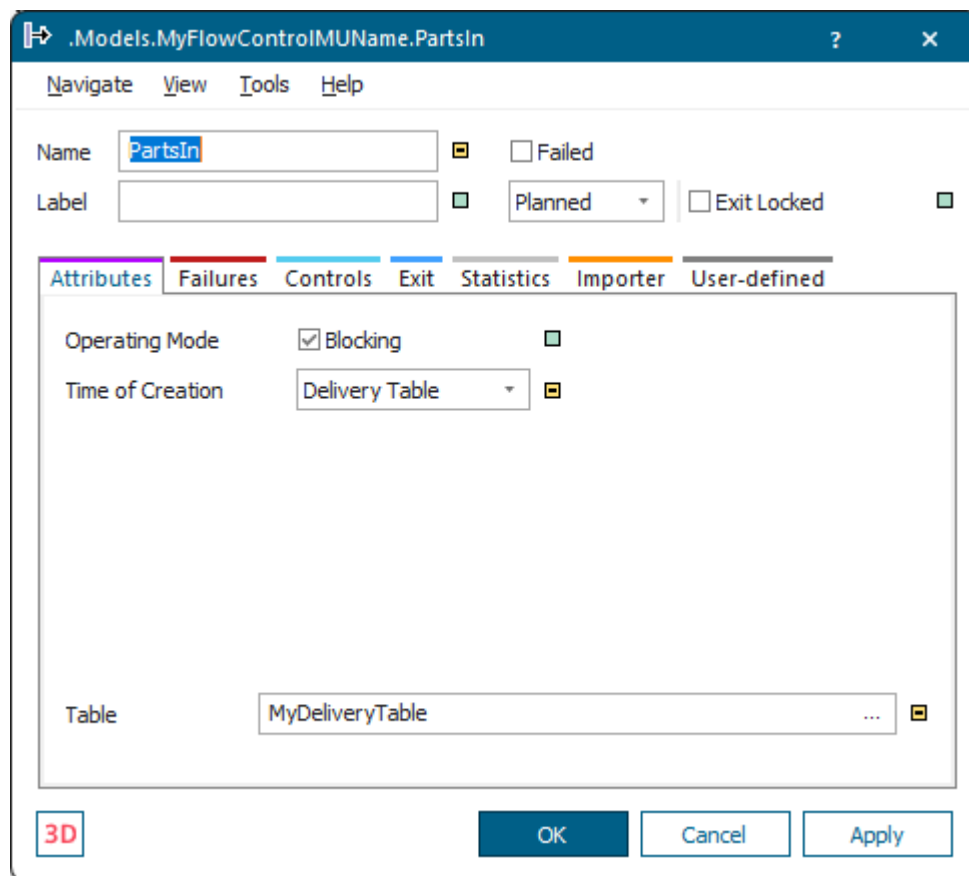


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Configure the Source that Produces the Parts [FlowControl]

Insert and configure the *Source* which produces the parts.

- Select when and how many parts of a certain type are to be produced.
We selected a **Delivery Table** in our example, as shown in the screenshot below.



- We then selected the name of the **Delivery Table**, which we inserted before, namely *MyDeliveryTable*. Selecting it automatically applied the correct format.
- Next, we entered the required information, compare the screenshot below. The *Source* which we named *PartsIn* in our sample model, produces six different parts with the names we entered into the column **Name**.

	time 1	object 2	integer 3	string 4	table 5
string	Delivery Time	MU	Number	Name	Attributes
1	0.0000	.MUs.MyPart		Red	MyAttr
2	0.0000	.MUs.MyPart		Yellow	MyAttr
3	0.0000	.MUs.MyPart		Green	MyAttr
4	0.0000	.MUs.MyPart		BlackRound	MyAttr
5	0.0000	.MUs.MyPart		BlackSquare	MyAttr
6	0.0000	.MUs.MyPart		BlackOctagonal	MyAttr
7					

To enter the attributes according to which the *FlowControl* distributes the parts among its successors, type in any name of your choice into the column **Attributes** to create a subtable with this name, and then double-click the name.

- For the part with the **Name** Red, we typed in MyAttr, double-clicked the cell, and entered Color is the **Name of the Attribute** and Red as the color of the part.

The screenshot shows the 'MyAttr' table with 8 rows. The first row has 'Red' in the 'Name' column. A context menu is open for this cell, showing a 'Color' table with 6 columns: string (Name of Attribute), integer, boolean, string, real, and time. The first row of the context menu has 'Color' in the 'Name of Attribute' column and 'Red' in the 'string' column.

	time 1	object 2	integer 3	string 4	table 5
string	Delivery Time	MU	Number	Name	Attributes
1	0.0000	.MUs.MyPart		Red	MyAttr
2	0.0000	.MUs.MyPart		Yellow	
3	0.0000	.MUs.MyPart		Green	
4	0.0000	.MUs.MyPart		BlackRound	
5	0.0000	.MUs.MyPart		BlackSquare	
6	0.0000	.MUs.MyPart		BlackOctagonal	
7					
8					

	string 1	integer 2	boolean 3	string 4	real 5	time 6
string	Name of Attribute					
1	Color			Red		
2						

- For the part with the **Name** BlackSquare, we typed in MyAttr, double-clicked the cell, and entered Color is the **Name of the Attribute** and Black as the color of the part. We also entered a second attribute, namely Shape, and typed in Square as its shape.

The screenshot shows the 'MyAttr' table with 12 rows. The fifth row has 'BlackSquare' in the 'Name' column. A context menu is open for this cell, showing a 'Color' table with 6 columns: string (Name of Attribute), integer, boolean, string, real, and time. The first row of the context menu has 'Color' in the 'Name of Attribute' column and 'Black' in the 'string' column. The second row has 'Shape' in the 'Name of Attribute' column and 'Square' in the 'string' column.

	time 1	object 2	integer 3	string 4	table 5
string	Delivery Time	MU	Number	Name	Attributes
1	0.0000	.MUs.MyPart		Red	MyAttr
2	0.0000	.MUs.MyPart		Yellow	MyAttr
3	0.0000	.MUs.MyPart		Green	MyAttr
4	0.0000	.MUs.MyPart		BlackRound	MyAttr
5	0.0000	.MUs.MyPart		BlackSquare	MyAttr
6	0.0000	.MUs.MyPart		BlackOctagonal	
7					
8					
9					
10					
11					
12					

	string 1	integer 2	boolean 3	string 4	real 5	time 6
string	Name of Attribute					
1	Color			Black		
2	Shape			Square		

- For the part with the **Name** BlackOctagonal, we typed in MyAttr, double-clicked the cell, and entered Color is the **Name of the Attribute** and Black as the color of the part. We also entered a second attribute, namely Shape, and typed in Octagonal as its shape.

Configure the Source that Produces the Parts [FlowControl]

.Models.MyFlowControlIMUName.MyDeliveryTable					
MyAttr					
	time	object	integer	string	table
string	1	2	3	4	5
	Delivery Time	MU	Number	Name	Attributes
1	0.0000	.MUs.MyPart		Red	MyAttr
2	0.0000	.MUs.MyPart		Yellow	MyAttr
3	0.0000	.MUs.MyPart		Green	MyAttr
4	0.0000	.MUs.MyPart		BlackRound	MyAttr
5	0.0000	.MUs.MyPart		BlackSquare	MyAttr
6	0.0000	.MUs.MyPart		BlackOctagonal	MyAttr
7					
8					
9					
10					
11					
12					
13					

.Models.MyFlowControlIMUName.MyDeliveryTable[5,6]					
Color					
	string	integer	boolean	string	real
string	1	2	3	4	5
	Name of Attribute				
1	Color			Black	
2	Shape			Octagonal	

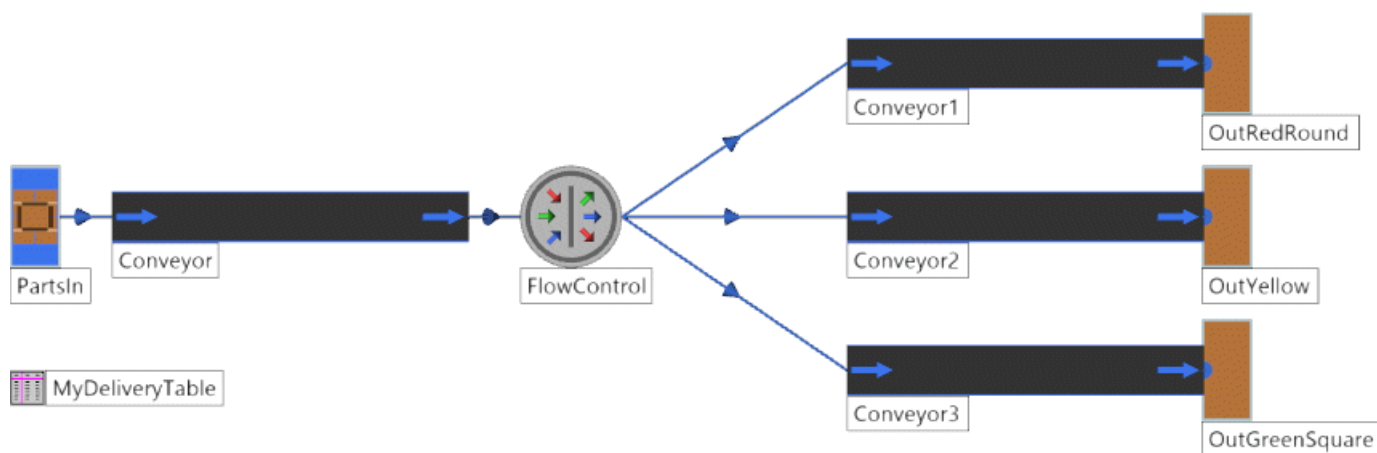
When the *Source* produces the parts, it creates the attributes, which we entered above, as *user-defined attributes* of data type *string* for the respective part.

.MUs.MyPart:1					
Name	Red	Number	1	Stopped	
Label		Conveying Direction	0 (forward)		
Attributes					
New...	Edit...	Delete			
Name	Value	Type	C..	I...	
Color	Red	string	*	-	

.MUs.MyPart:4					
Name	BlackRound	Number	4	Stopped	
Label		Conveying Direction	0 (forward)		
Attributes					
New...	Edit...	Delete			
Name	Value	Type	C..	I...	
Color	Black	string	*	-	
Shape	Round	string	*	-	

.MUs.MyPart:6					
Name	BlackOctagonal	Number	6	Stopped	
Label		Conveying Direction	0 (forward)		
Attributes					
New...	Edit...	Delete			
Name	Value	Type	C..	I...	
Color	Black	string	*	-	
Shape	Octagonal	string	*	-	

Then, insert the other objects, so that they match the screenshot below.

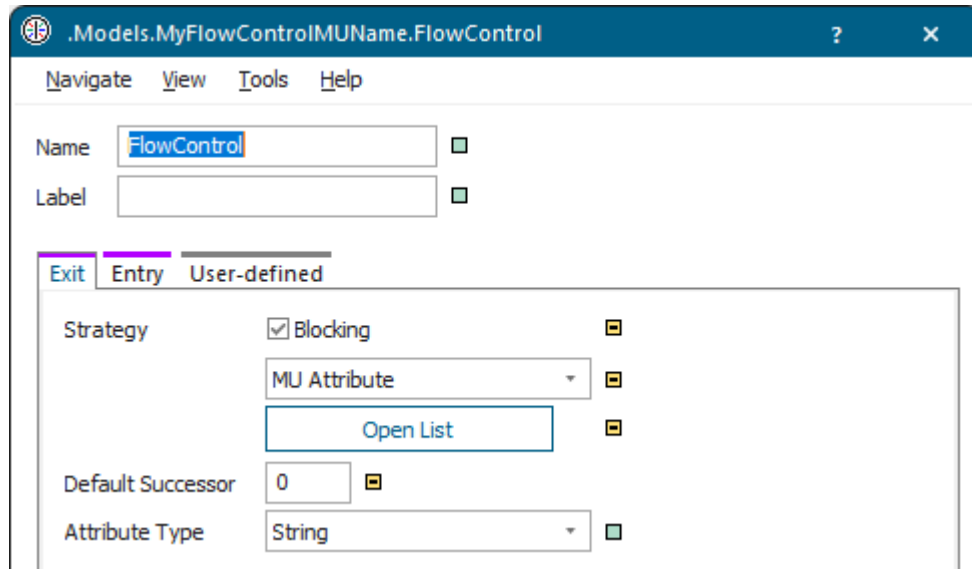


Back to [Distribute Parts Among its Successors with the FlowControl](#)

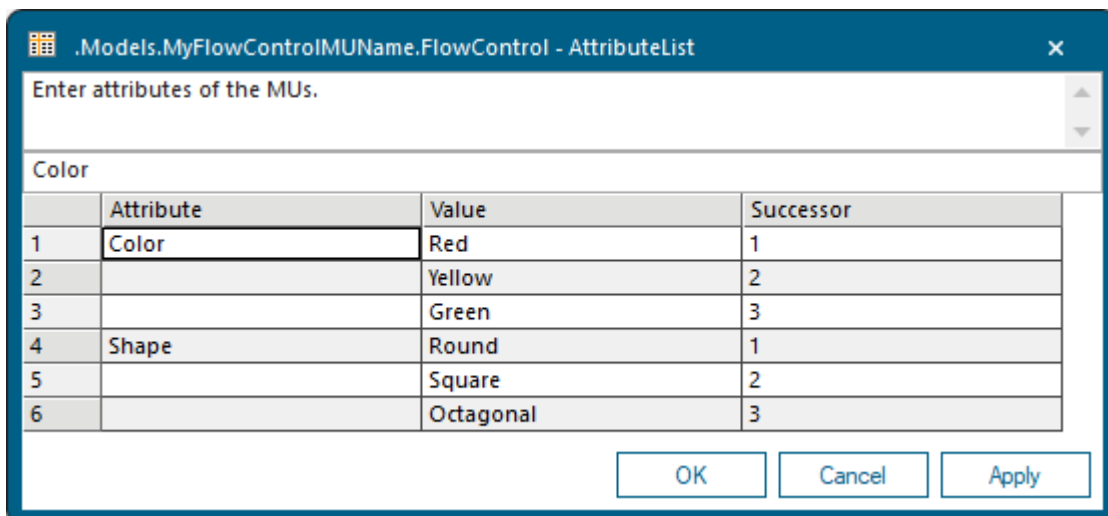
Configure the FlowControl Which Distributes the Parts

Insert and configure the *FlowControl* which distributes the parts among its successors in the flow of materials:

- As we want the *FlowControl* to distribute the parts according to the *user-defined attributes*, which we defined above, select the setting **MU Attribute** from the drop-down list **Strategy**.

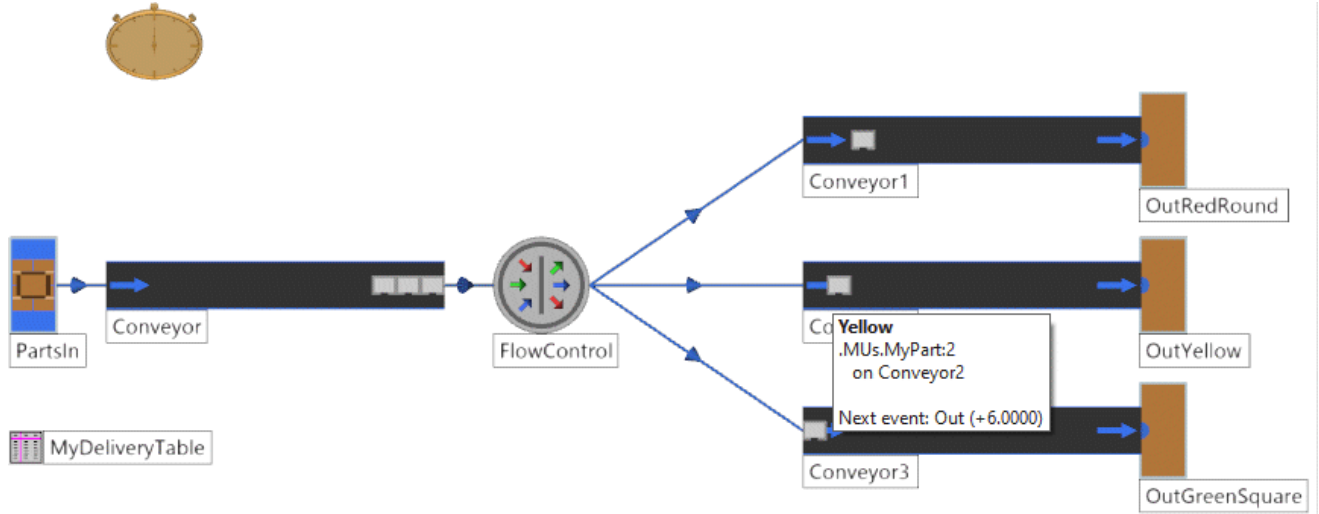


- Then, click **Open List** to open the table, into which you have to type in the names of the **attributes**, their **values**, and the numbers of the **successors**. The *FlowControl* searches the table from top to bottom until it finds an attribute with the value you typed in. The *FlowControl* then moves the *part* on to that successor.

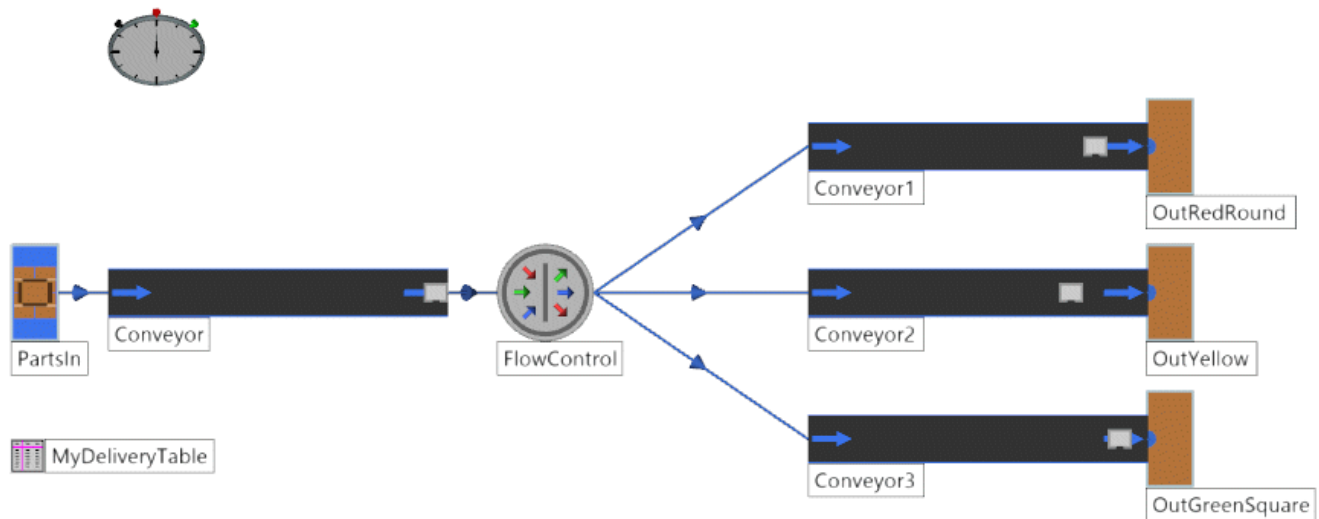


- The number of the **default successor** decides to which successor the *FlowControl* sends the part whose attributes do not meet the criteria in the list. In our example neither the attribute **Color** for the attribute **Shape** of the part *BlackOctagonal* meets the criteria.
- Run the simulation and watch what happens:

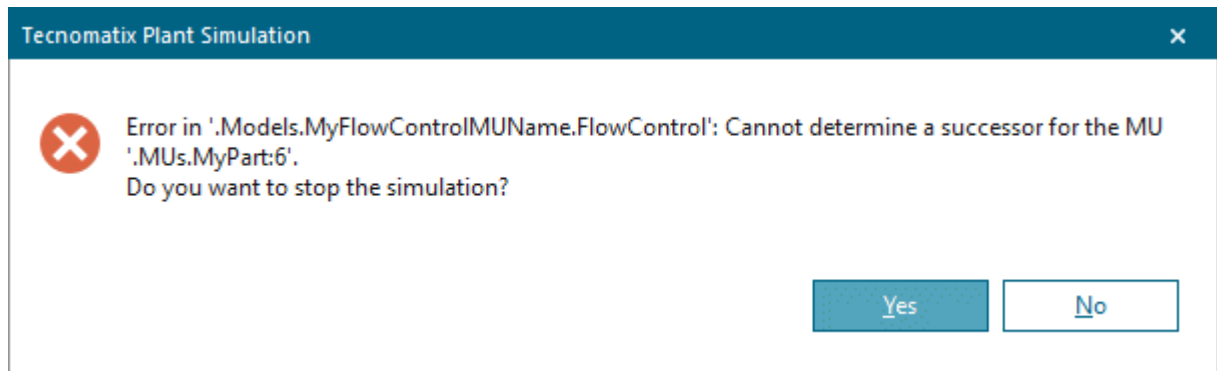
- When you use the default setting **1**, the *FlowControl* moves the part on to the successor object with the number 1.



- When you type in **0**, the *FlowControl* does not move the part on. It remains in front of the *FlowControl*, thus blocking the following parts.



- When you type in **-1**, or any other negative number, *Plant Simulation* show an error message. You can then react accordingly by determining what should happen to the part.

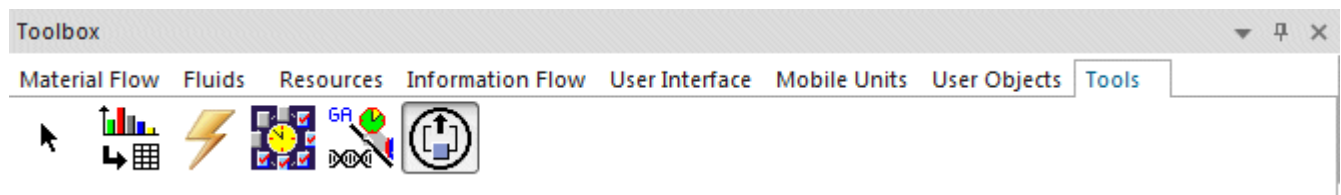


Back to [Distribute Parts Among its Successors with the FlowControl](#)

Loading, Unloading, and Reloading Parts

Load, Unload, and Reload Parts with the TransferStation

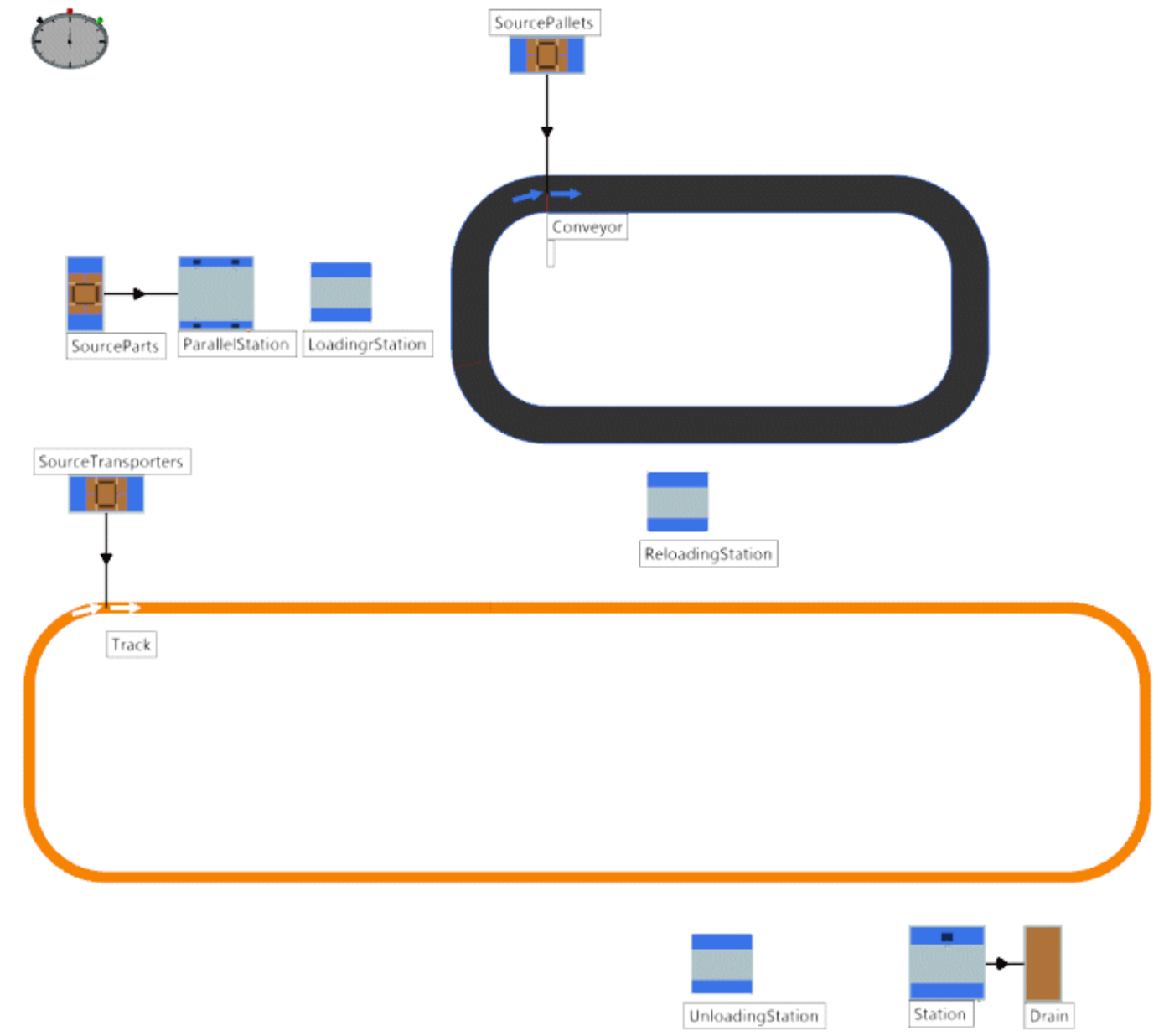
To define your own way of transferring parts from station to station, you can use the [TransferStation](#). You can insert the *TransferStation* into your simulation model from the folder **Tools** in the *Class Library* or from the toolbar **Tools** in the *Toolbox*.



Note

The *TransferStation* is, by default, not shown when you create a new model. The [PickAndPlace robot](#) can emulate most of the functions of the *TransferStation*.

In our example below we load a block of four parts onto a pallet, which is transported to another section within the plant on a conveyor line. At the location in the plant, which is marked by a sensor, we unload parts in blocks of two from the pallets and reload them onto a transport vehicle. Finally, we unload parts in blocks of one from the transport vehicle onto a processing station and then remove them from the plant.



Back to [Transfer Parts from Station to Station](#)

Load Parts with the Transfer Station

One of your most basic tasks when creating a simulation model will be to load parts onto a means of transport, a *Container*  or a *Transporter*  and to unload them from a means of transport.

When **loading** parts, the *TransferStation* gets these parts from the parts station and loads them onto a means of transport, which has to be located at the target station. The loading process starts as soon as part and means of transport are ready.

To model a loading process, proceed like this:

- Insert the conveyor named *Conveyor*, which transports the parts.

Parts from:		Target is on:	
ParallelStation	...	Conveyor	...
Sensor position:	Lane A	Sensor position:	0

- Insert a *Source*, which creates the parts. In our sample model we named it *SourceParts*.

.Models.Model.SourceParts

Navigate View Tools Help

Name: SourceParts ☐ Failed

Label: ☐ Planned ☐ Exit Locked

Attributes Failures Controls Exit Statistics Importer User-defined

Operating Mode ☒ Blocking ☐

Time of Creation Interval Adjustable ☐ Amount -1 ☐

Interval Const 0 ☐

Start Const 0 ☐

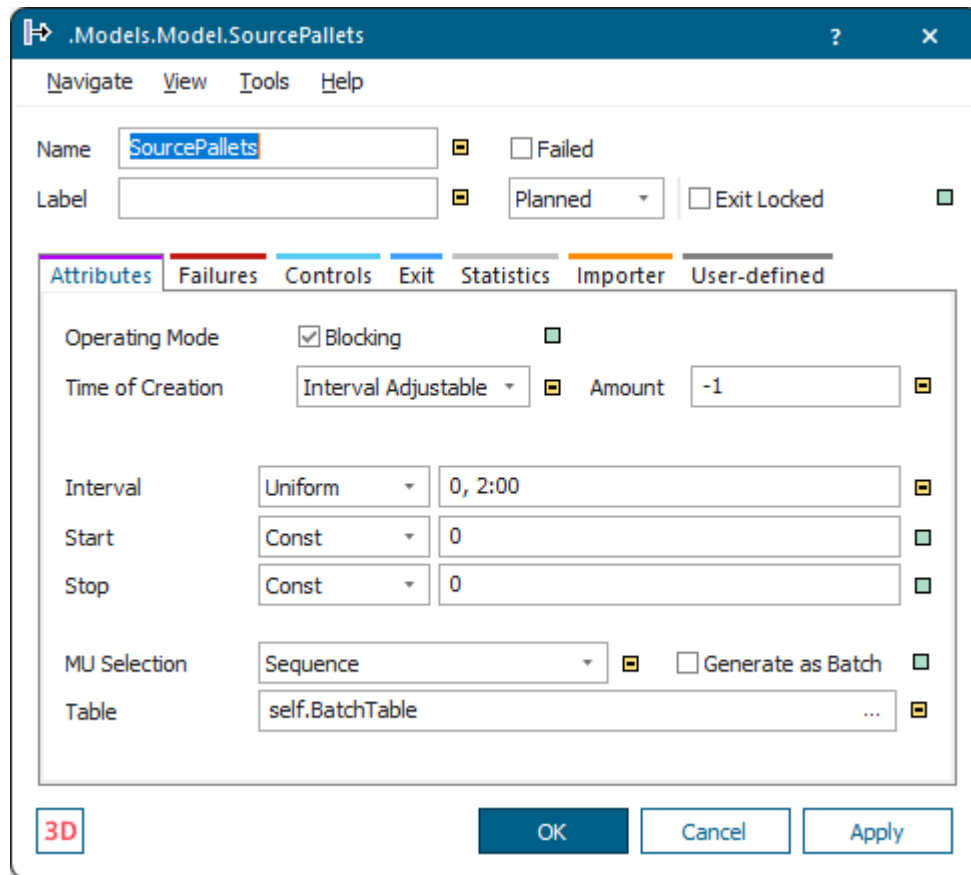
Stop Const 0 ☐

MU Selection Constant ☐

MU .MUs.Part ☐

3D OK Cancel Apply

- Insert a *ParallelStation*, which processes the parts before they are loaded onto pallets.
- Insert a *Source*, which creates the pallets onto which the *TransferStation* loads the parts. In our sample model we named it *SourcePallets*.



- Insert a *TransferStation* , which loads the processed parts onto the *Container* , i.e., the means of transport, which moves on the conveyor. In our sample model we named it *LoadingStation*. Then, we have to tell this station what we want it to do:
 - We want to load parts, so we select **Load** as the **Station type**.
 - Then, we have to tell it where the **parts** to be loaded come from. For this we click the  button and select the *ParallelStation* in the dialog **Select Object**.
 - Next, we have to tell it where the **means of transport**, our *Container*, is **located**. For this we click the  button and select the *Conveyor* in the dialog **Select Object**.
 - Finally, we have to designate the position on the *Conveyor*, where the *LoadingStation* loads the parts onto the pallets. For this we type 0 meters as the **position of the sensor** into the text box on the tab **Attributes**. The *LoadingStation* automatically inserts this sensor into the object *Conveyor*.

To show a *tooltip* with information about the sensor, drag the mouse over the red line designating it on the object.

Note

To successfully transfer the *MU* onto the target station, you have to make sure that the target station provides enough space to accommodate the incoming *MU*.

The screenshot shows the **.Models.Frame.LoadingStation** dialog box. The **Name** field is set to `loadingrStation` and the **Label** field is empty. The **Planned** checkbox is checked. The **Attributes** tab is selected. In the **Station type** section, the **Load** radio button is selected. The **Parts from:** dropdown is set to `ParallelStation` and the **Target is on:** dropdown is set to `Conveyor`. The **Sensor position:** dropdown is set to `Lane A` and the **Sensor position:** text field is set to `0`. The **OK**, **Cancel**, and **Apply** buttons are at the bottom.

To set how we want the parts to be loaded, we click the tab **Advanced Attributes**.

We want to load **all blocks** of parts in blocks of 4.

The screenshot shows the **.Models.Frame.LoadingStation** dialog box with the **Advanced Attributes** tab selected. The **Name** field is set to `LoadingStation` and the **Label** field is empty. The **Planned** checkbox is checked. In the **Load/unload** section, the **all** radio button is selected, the **-1** text field is set to `2`, and the **blocks** dropdown is set to `2`. The **Block size** text field is set to `2`. The **Specific type** checkbox is unchecked, the **Open Table** button is visible, and the **Homogeneous loading** checkbox is unchecked. The **Always stop target/container** checkbox is checked. In the **Behavior for failures:** section, the **Stop container on part station** and **Stop container on target station** checkboxes are both checked. The **OK**, **Cancel**, and **Apply** buttons are at the bottom.

Compare Create a Sensor [material flow objects]

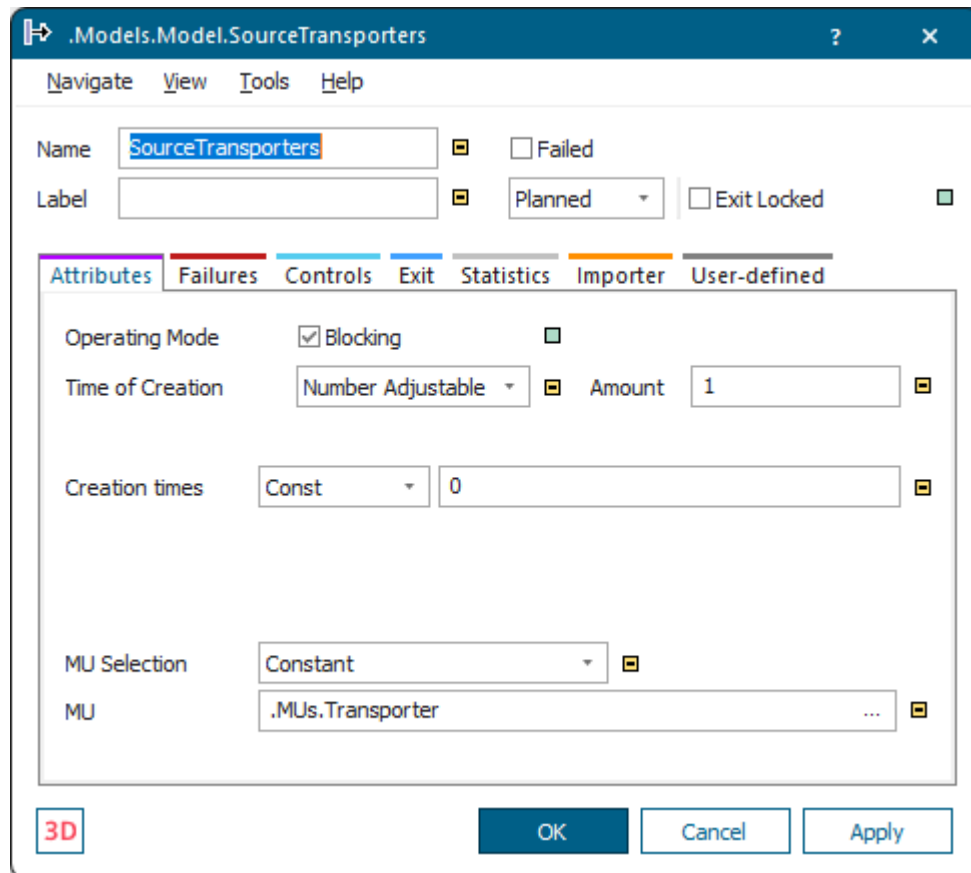
Back to Load, Unload, and Reload Parts with the TransferStation

Reload Parts with the Transfer Station

When **reloading** parts, the *TransferStation* takes these parts from a means of transport (*Container* or *Transporter*), which is located at the parts station, and places them onto a *Container* or a *Transporter*, which is located at the target station. The parts station is the station, which provides the parts to be transported. The target station is the station onto which the parts will be placed. The reloading process starts as soon as both means of transport are ready.

To model a reloading process, proceed like this:

- Insert a *Source*, which creates the *Transporters*. In our sample model we named it *SourceTransporters*.

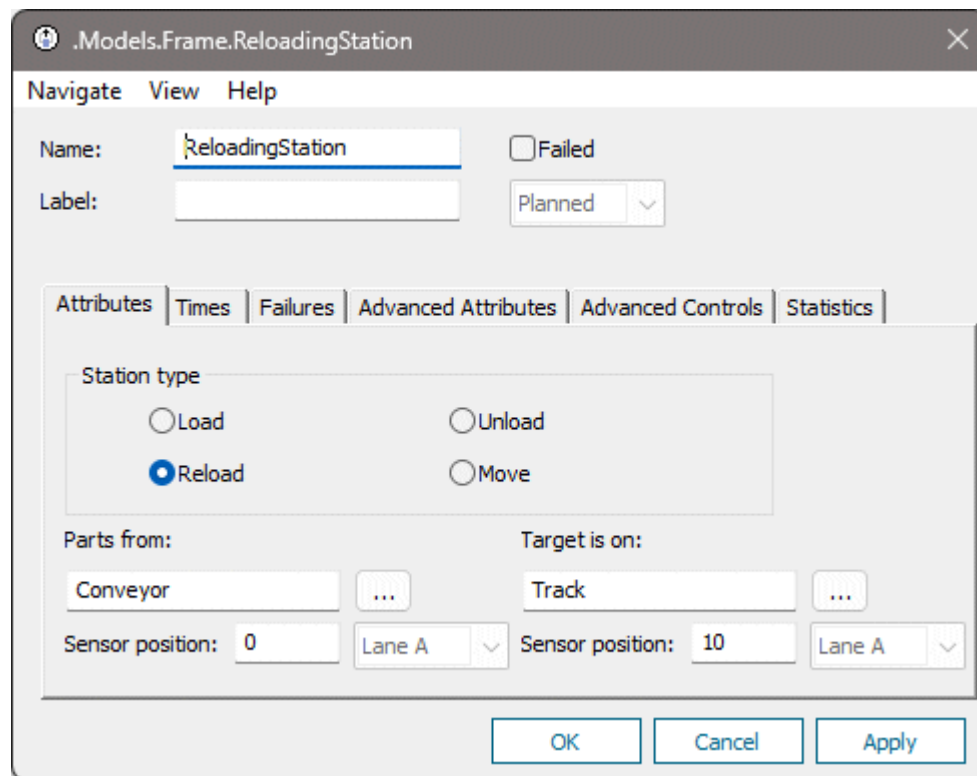


- Insert the *Track*, on which the *Transporter* transports there loaded parts.
- Insert a *TransferStation*, which reloads the transported parts from the *Container* into the *Transporter*. In our sample model we named it *ReloadingStation*. Then, we tell this station what we want it to do.

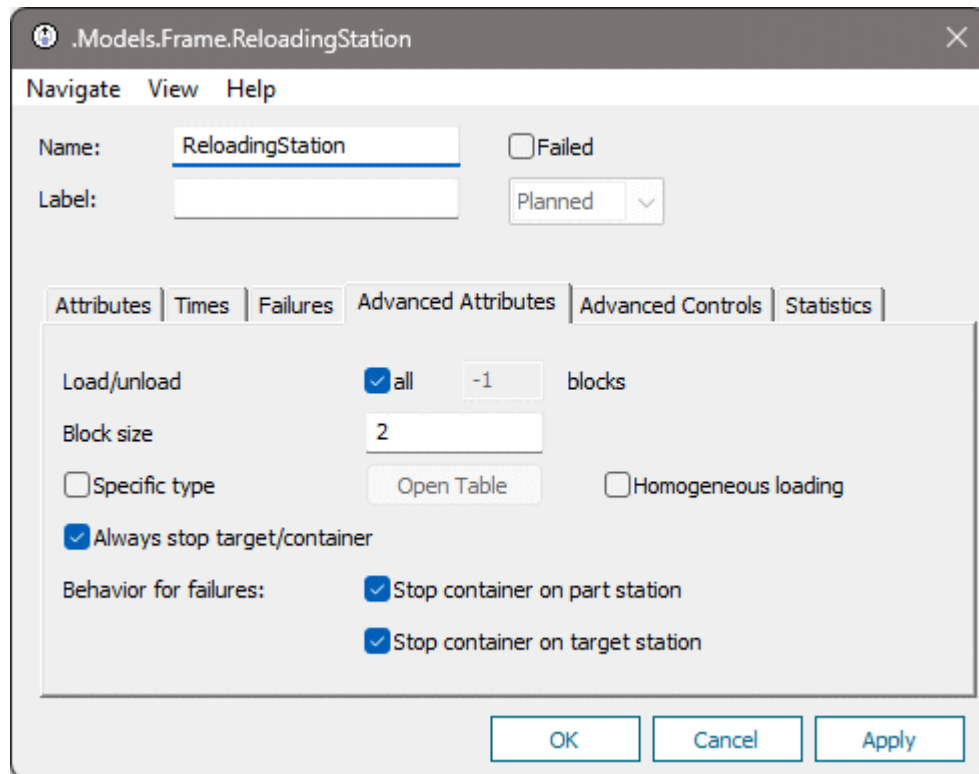
- We want to reload parts, so we select **Reload** as the **Station type**.
- Then, we have to tell it where the **parts** to be reloaded **come from**. For this we drag the *Line/Conveyor* from the *Frame* over the text box in the dialog of the *ReloadingStation* and drop it there.
- Next, we have to tell it where the **means of transport**, our *Transporter*, **is located**. For this we drag the *Track* from the *Frame* over the text box in the dialog of the *ReloadingStation* and drop it there.
- Finally, we have to designate the exact spot on the *Line/Conveyor*, where the *ReloadingStation* removes the parts from the pallets. For this we enter 22 meters as the **position of the sensor**, which the *ReloadingStation* automatically enters into the object *Line/Conveyor*.

Note

To successfully transfer the *MU* into the target station, you have to make sure that the target station provides enough space to accommodate the incoming *MU*.



- As we want to reload the parts from the pallet onto a *Transporter*, we have to designate the exact spot on the *Track*, where the *ReloadingStation* loads the parts onto the *Transporter*. For this we enter 15 meters as the **position of the sensor**, which the *ReloadingStation* automatically enters into the object *Track*.
- To set how we want the parts to be reloaded, we click the tab **Advanced Attributes**. We want to load parts in blocks of 2.



Compare Create Sensors

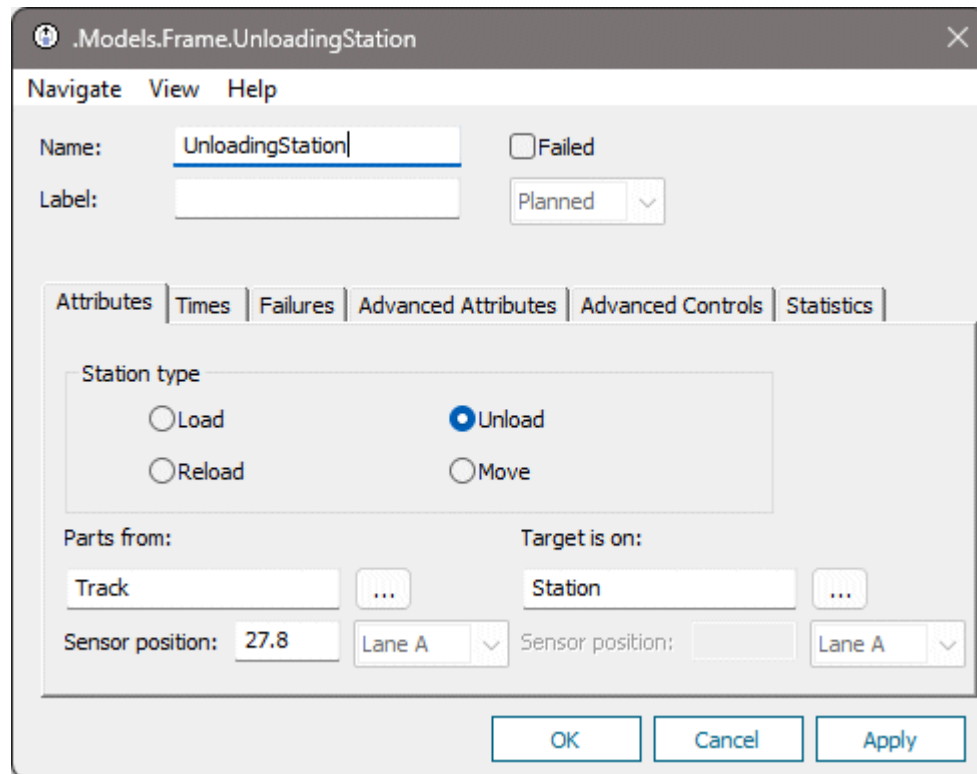
Back to Load, Unload, and Reload Parts with the TransferStation

Unload Parts with the Transfer Station

When **unloading** parts, the *TransferStation* unloads these parts from a *Container* or a *Transporter*, which is located at the parts station, and places them onto the target station. The unloading process starts as soon as a means of transport is ready at the parts station.

To model an unloading process, proceed like this:

- Insert a *Station*, on to which the *UnloadingStation* unloads the parts from the *Transporters*.
- Insert a *Drain*, which removes the parts from the plant.
- Insert a *TransferStation*, which unloads the parts from the *Transporter* and places them on to the processing station named *Station*. In our sample model we named it *UnloadingStation*. Now we tell this station what we want it to do.
 - We want to unload parts, so we select **Unload** as the **Station type**.



- Then, we have to tell it where the **parts** to be unloaded **come from**. For this we drag the *Track* from the *Frame* over the text box in the dialog of the *UnloadingStation* and drop it there.
- Next, we have to tell it where the **means of transport**, our *Transporter*, **is located**. For this we drag the *Station* from the *Frame* over the text box in the dialog of the *UnloadingStation* and drop it there.
- To designate the exact spot on the *Track* , where the *UnloadingStation* unloads the parts from the *Transporters*  in to the processing station *Station*, we enter 43 meters as the sensor position. The *UnloadingStation* automatically enters this sensor into the object *Track*.
- To set how we want the parts to be unloaded, we click the tab **Advanced Attributes**. We want to unload **all blocks** of parts in blocks of 1.
- Insert a *Method*, enter `deleteMovables` is the source code, and name the *Method* *reset*.
- Insert an *EventController* into the *Frame*, and run the simulation.

You can then change a number of settings, such as the dimensions of the *Transporters* and the *Containers*, the block size, the number of blocks, etc. and watch how these changes affect the simulation model.

Compare Create a Sensor [material flow objects]

Back to Load, Unload, and Reload Parts with the TransferStation